GAMMA-RAY BURST CLASSIFICATION

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- Github repository

T_{90} DISTRIBUTION

- **Gamma-ray bursts** (GRBs) are usually classified into two different categories, according to their duration: **short GRBs** and **long GRBs**. The distribution of the burst duration, T_{90}, is usually modelled as the **weighted sum of two log-normal distributions**:

$$p(T_{90}) = w_1\, \mathcal{N}(logT_{90} | \mu_1, \sigma_1^2) + (1 - w_1)\mathcal{N}(logT_{90} | \mu_2, \sigma_2^2)$$

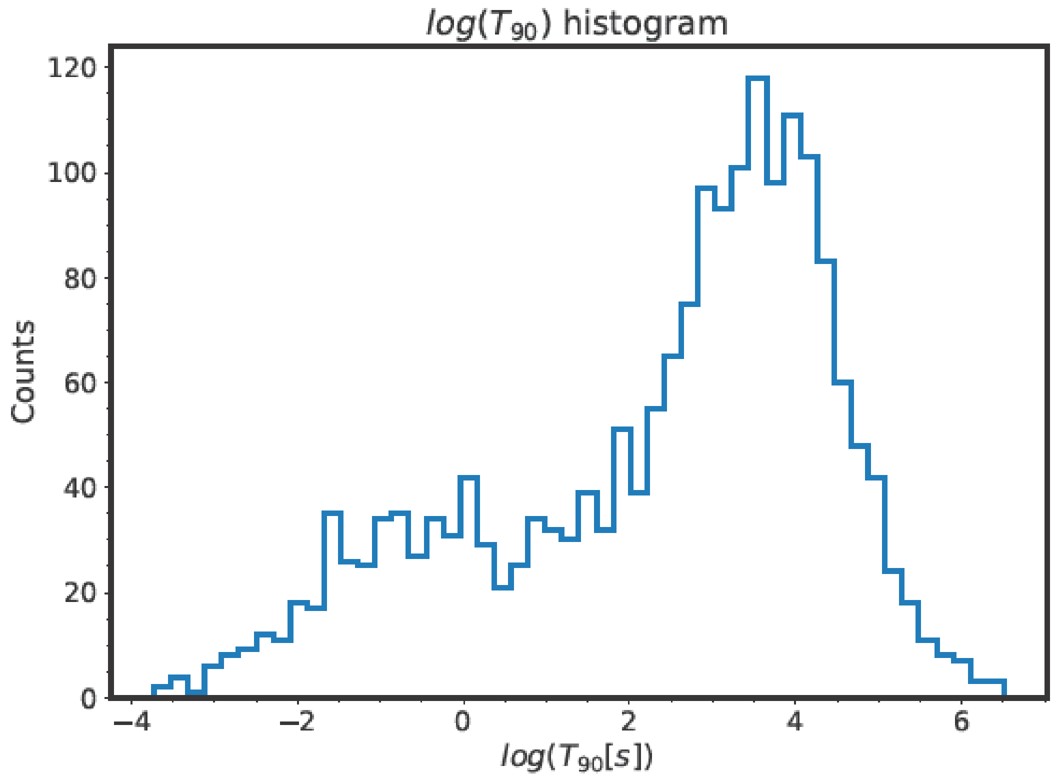
- The available data are $x = \log T_{90}$; the distribution becomes a **weighted sum of two normal distributions:**

$$p(\mathrm{x}) = w_1\, \mathcal{N}(\mathrm{x} | \mu_1, \sigma_1^2) + (1 - w_1)\mathcal{N}(\mathrm{x} | \mu_2, \sigma_2^2)$$

$$p(T_{90}) = p(logT_{90})\frac{d(logT_{90})}{dT_{90}} = p(logT_{90})/T_{90} \quad \Rightarrow \quad \mathrm{p(x)} = p(logT_{90}) = T_{90}\,p(T_{90})$$

PROBLEM 1A – PARAMETER ESTIMATION

- **Problem 1a:** estimate the distribution parameters while ignoring measurement uncertainties.



METROPOLIS ALGORITHM

The goal is to draw samples from a target distribution f(x); the algorithm is:

- Choose starting state x_0.
- From current x_0 draw x' from a symmetric proposal (for example Gaussian).
- Compute the acceptance $a = min\left(1, \frac{f(x')}{f(x_0)}\right)$.
- With probability a set x_1 = x', otherwise $x_1 = x_0$.
- Repeat points 2-4 to build a **Markov chain** whose **stationary distribution** is f(x).

Sequence of random states where each new state depends only on the current one

Probability distribution that remains unchanged as the chain evolves (after burnin)

BAYES THEOREM

- $\boldsymbol{x} = \{logT_{90_i}\}_{i=1}^{N}$
- $\boldsymbol{\theta} = (w_1, \mu_1, \delta_{12}, \sigma_1, \sigma_2) \qquad \delta_{12} \stackrel{\text{def}}{=} \mu_2 - \mu_1$
- I = background information (the term will be omitted in later equations for simplicity)

$$p(\boldsymbol{\theta}|\boldsymbol{x}, I) = \frac{p(\boldsymbol{x}|\boldsymbol{\theta}, I)p(\boldsymbol{\theta}|I)}{p(\boldsymbol{x}|I)} \propto \mathrm{p}(\boldsymbol{\theta}|I)\prod_{i=1}^{N} p(x_i|\boldsymbol{\theta}, I)$$

- Data x_i are assumed independent, and each $p(x_i|\boldsymbol{\theta}, I)$ is a mixture of two Gaussian components
- $p(\boldsymbol{\theta}|I) = p(w_1)p(\mu_1)p(\delta_{12})p(\sigma_1)p(\sigma_2)$

PRIOR

- $p(w_1) = Beta(2,2) \qquad Beta(\alpha, \beta) \propto x^{\alpha-1}(1-x)^{\beta-1}$

A Beta(2,2) prior assigns zero density at the boundaries 0 and 1, discouraging extreme values for w_1 when only one Gaussian is present.

- $p(\mu_1) \propto \mathcal{U}(-4,7)$
- $p(\delta_{12}) \propto \mathcal{U}(0,10)$

δ_{12} prevents label switching by using the ordering constraint : $\mu_2 \geq \mu_1$

- $p(log\sigma_i) \propto \mathcal{U}\big(log(0.1), log(6)\big) \quad \Rightarrow \quad p(\sigma_i) \propto \frac{1}{\sigma_i}$

A uniform prior on logσ ensures **scale invariance**:

$$\sigma' = a\sigma \quad \Rightarrow \quad p(\sigma')d\sigma' = \frac{1}{\sigma'}d\sigma' = \frac{1}{a\sigma}ad\sigma = \frac{1}{\sigma}d\sigma = p(\sigma)d\sigma$$

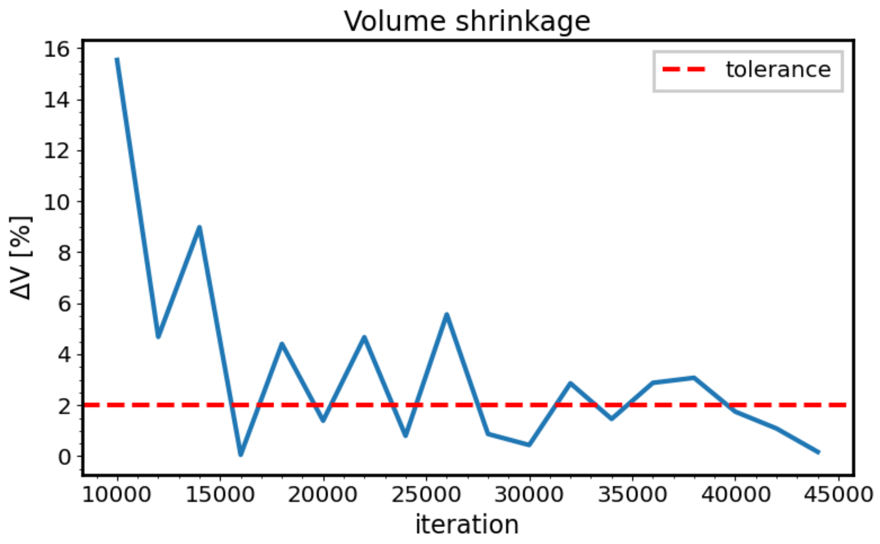
ADAPTIVE PROPOSAL

- The proposal is a multivariate Gaussian.
- If the covariance is $I_{5\times5}$ or $0.01 \times I_{5\times5}$, the autocorrelation decays to 0 after lag 1000.

$$\rho(\tau) = \frac{\mathbb{E}[(x(t) - \bar{x})(x(t+\tau) - \bar{x})]}{\mathbb{E}[(x(t) - \bar{x})^2]}$$

- Two runs:
  1. **Warm-up**: adapt the proposal (samples not valid, chain is non-Markovian).
  2. **Frozen**: draw samples using the final warm-up proposal.
- During the warm-up run it is not possible to collect valid samples because the proposal is updated based on all previous iterations, so **the chain is not Markovian**.
- Covariance update: $\Sigma = sC + \varepsilon I$, with fixed scale $s = 1$ and small ε to prevent divergence.

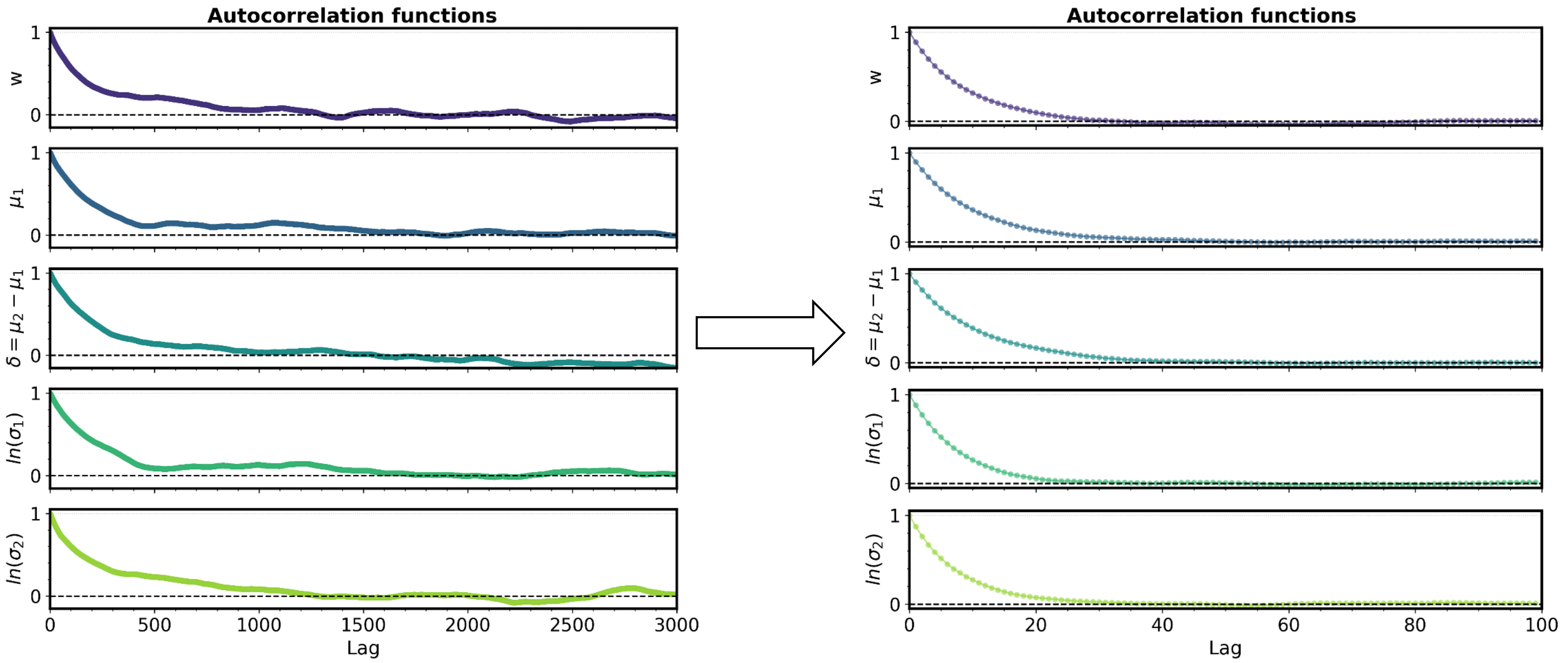
ADAPTIVE COVARIANCE TUNING (WARM-UP)



- With $\Sigma = 0.01 \times I_{5\times5}$, start a Metropolis algorithm.
- Every *adapt_interval* steps (2000), compute:

$$\Sigma_{new} = Cov\left(samples_{burnin:}\right) + \varepsilon I$$

  and set $\Sigma = \Sigma_{new}$.
- **Stopping criterion:** adaptation stops early if all conditions are met:
  1. Minimum number of iterations (*min_adapt=10000*) has passed.
  2. Log-determinant change is small: $|\boldsymbol{\Delta logdet}(\boldsymbol{\Sigma})| < \boldsymbol{tol\_logdet} = 2 \times 10^{-2}$
  3. This small change persists for several consecutive checks (*consecutive_stable=3*) to avoid transient fluctuations.



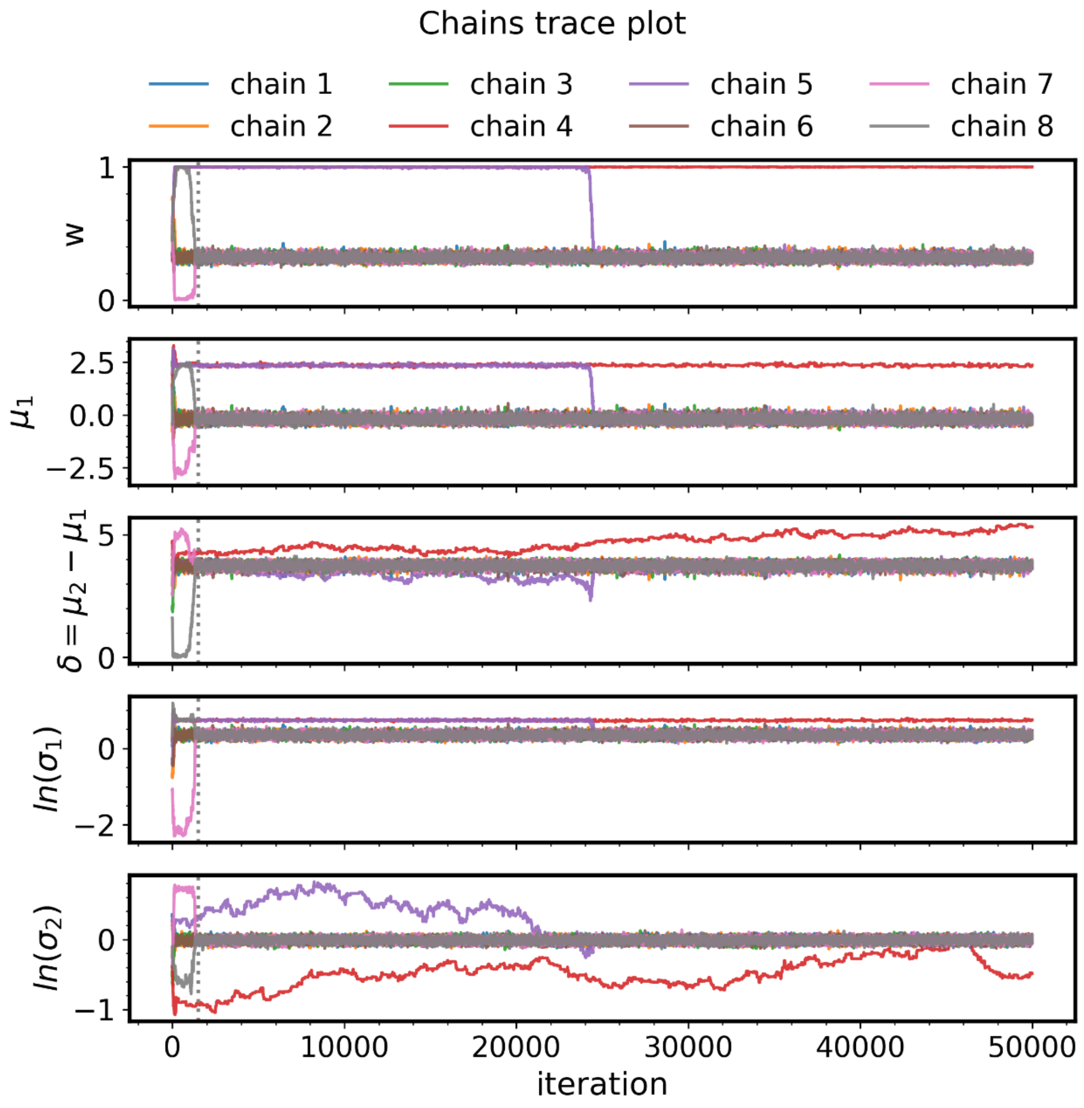
- **Acceptance rate**: 0.0044 → 0.3126
- The data in the plots are **generated** from a mixturee of Gaussians.
- The whole adaptive–proposal mechanism **is not needed**: using the sample covariance from the first (non-adaptive) run yields results that are practically identical.

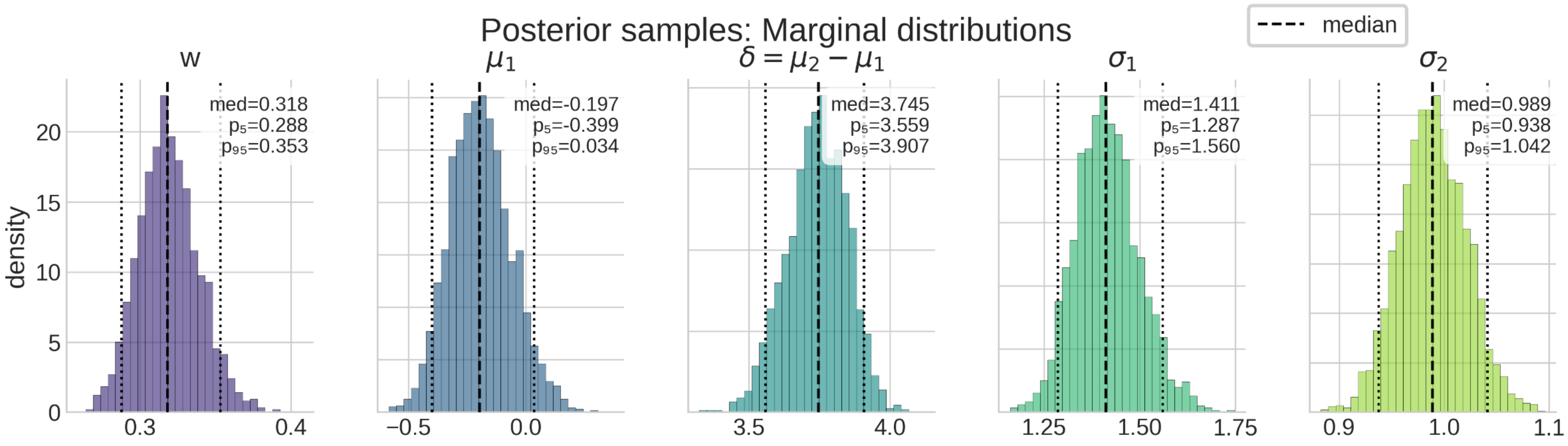
INITIAL STATE

- The chain can get stuck near w = 0 or w = 1.
- Chains get stuck at **boundary peaks** because local proposals land in much lower posterior density and are therefore almost always rejected.
- Peaks at $w \approx 0$ or $w \approx 1$ arise because there exist parameter configurations in which one mixture component fits the data by itself, making the other component effectively irrelevant.
- To minimize the risk, the chain was initialized at w = 0.5
- Other parameters were initialized to:
  - $\sigma_i = 1$
  - $\mu_1 = p_{25}$ (p_i = i-th percentile)
  - $\delta_{12} = p_{75} - p_{25}$

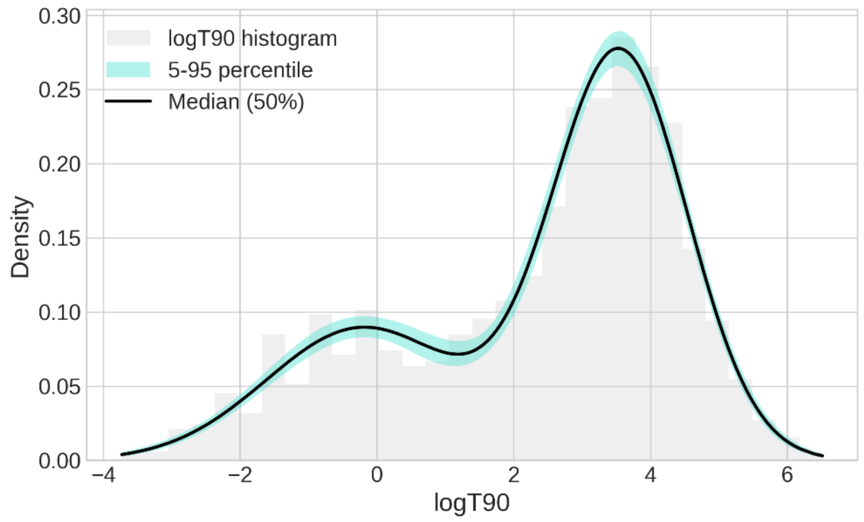
ENSEMBLE

- Initial state validated on simulated data; real data stability uncertain.
- Ensemble of **8 chains** with **different initial states**.
- Initial states obtained by **Gaussian perturbation** (mean = base init; std = 10% of parameter range; clipped to prior boundaries).
- 2/8 chains stuck at $w \approx 1 \rightarrow$ discarded.
- **Thinning**: first lag where autocorrelation < 0 (max across parameters).
- **Burnin** = 1500.





Mean acceptace rate = 0.269
#samples = 2975



PARAMETER VALUES

Parameters	Mean ± std	Median±$[p_{16}, p_{84}]$(68%)	$[p_5, p_{95}]$ (90%)
w_1	0.319 ± 0.020	$0.318^{+0.021}_{-0.019}$	[0.288, 0.353]
μ_1 [log s]	-0.19 ± 0.13	$-0.20^{+0.14}_{-0.12}$	[-0.40, 0.03]
δ_{12} [log s]	3.74 ± 0.11	$3.75^{+0.10}_{-0.11}$	[3.56, 3.91]
μ_2 [log s]	3.548 ± 0.045	$3.548^{+0.044}_{-0.044}$	[3.472, 3.621]
σ_1 [log s]	1.415 ± 0.084	$1.411^{+0.087}_{-0.079}$	[1.287, 1.560]
σ_2 [log s]	0.989 ± 0.032	$0.989^{+0.032}_{-0.031}$	[0.938, 1.042]

PROBLEM 1B - GAUSSIAN UNCERTAINTIES

- **Problem 1b:** estimate the distribution parameters assuming **Gaussian uncertainties** on each *logT90.*

$$x_{obs} = x + \varepsilon \qquad \varepsilon \sim \mathcal{N}\left(0, \sigma_{obs}^2\right)$$

- Marginal in x:

$$p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \int_{-\infty}^{+\infty} p(x_{obs}|x, \boldsymbol{\theta}, \sigma_{obs}) p(x|\boldsymbol{\theta}, \sigma_{obs}) dx$$

- The second term in the integral is indipendent of σ_{obs}. Let us assume only **one component**.

$$p(x|\boldsymbol{\theta}, \sigma_{obs}) = p(x|\boldsymbol{\theta}) = \mathcal{N}\left(x \middle| \mu_1, \sigma_1^2\right)$$

- The first term in the integral is indipendent of $\boldsymbol{\theta}$*.*

$$p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \int_{-\infty}^{+\infty} \mathcal{N}\left(x_{obs} \middle| x, \sigma_{obs}^2\right) \mathcal{N}\left(x \middle| \mu_1, \sigma_1^2\right) dx$$

METHOD 1: COMPLETING THE SQUARE

- $p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \frac{1}{2\pi\sigma\sigma_{obs}} \int_{-\infty}^{+\infty} exp\left[-\frac{1}{2}\left(\frac{(x_{obs}-x)^2}{\sigma_{obs}^2} + \frac{(x-\mu_1)^2}{\sigma_1^2}\right)\right] dx$

- $\begin{cases} A = 1/\sigma_{obs}^2 + 1/\sigma_1^2 \\ B = x_{obs}/\sigma_{obs}^2 + \mu_1/\sigma_1^2 \\ C = x_{obs}^2/\sigma_{obs}^2 + \mu_1^2/\sigma_1^2 \end{cases} \implies \frac{(x_{obs}-x)^2}{\sigma_{obs}^2} + \frac{(x-\mu_1)^2}{\sigma_1^2} = Ax^2 - 2Bx + C = A\left(x - \frac{B}{A}\right)^2 + C - \frac{B^2}{A}$

- $p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \frac{1}{2\pi\sigma\sigma_{obs}} exp\left[-\frac{1}{2}\left(C - \frac{B^2}{A}\right)\right] \underbrace{\int_{-\infty}^{+\infty} exp\left[-\frac{1}{2}A\left(x - \frac{B}{A}\right)^2\right] dx}_{= \sqrt{2\pi/A}}$

- Replacing A, B, and C:

$$\implies \quad p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_{obs}^2)}} exp\left[-\frac{1}{2}\frac{(x_{obs} - \mu_1)^2}{\sigma_1^2 + \sigma_{obs}^2}\right] = \mathcal{N}\left(x_{obs}|\mu_1, \sigma_1^2 + \sigma_{obs}^2\right)$$

METHOD 2: FOURIER TRASFORM

- $p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \int_{-\infty}^{+\infty} g(x_{obs} - x) f(x) dx = (f * g)(x_{obs}) = \mathcal{F}^{-1}\left(\tilde{g}(k)\tilde{f}(k)\right)$

- $\mathcal{F}\left(e^{-ax^2}\right) = \sqrt{\frac{\pi}{a}} e^{-\frac{k^2}{4a}} \quad \Rightarrow \quad \begin{cases} \tilde{\mathrm{g}}(k) = \mathcal{F}\left(\mathrm{g}(\mathrm{x})\right) = \mathcal{F}\left(\frac{1}{\sqrt{2\pi\sigma_{obs}^2}} e^{-\frac{x^2}{2\sigma_{\mathrm{obs}}^2}}\right) = e^{-\frac{1}{2}\sigma_{\mathrm{obs}}^2 k^2} \\ \tilde{f}(k) = \mathcal{F}\left(f(x-\mu_1)\right) = e^{-i\mu_1 k}\mathcal{F}\left(\frac{1}{\sqrt{2\pi\sigma_1^2}} e^{-\frac{x^2}{2\sigma_1^2}}\right) = e^{-i\mu_1 k} e^{-\frac{1}{2}\sigma_1^2 k^2} \end{cases}$

- $\tilde{f}(k)\tilde{g}(k) = e^{-\frac{1}{2}(\sigma_1^2+\sigma_{obs}^2)k^2} e^{-i\mu_1 k} \stackrel{\text{def}}{=} \tilde{h}(k) e^{-i\mu_1 k}$

- $\mathcal{F}^{-1}\left(\tilde{h}(k)\right) = \frac{1}{\sqrt{2\pi\left(\sigma_1^2+\sigma_{obs}^2\right)}} e^{-\frac{1}{2}\frac{x_{obs}^2}{\sigma_1^2+\sigma_{obs}^2}}$

$$\Longrightarrow \quad \mathcal{F}^{-1}\left(\tilde{f}(k)\tilde{g}(k)\right) = h(x_{obs} - \mu_1) = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_{obs}^2)}} e^{-\frac{1}{2}\frac{(x_{obs}-\mu_1)^2}{\sigma_1^2+\sigma_{obs}^2}} = \mathcal{N}\left(x_{obs}|\mu_1, \sigma_1^2 + \sigma_{obs}^2\right)$$

GAUSSIAN MIXTURE GENERALIZATION

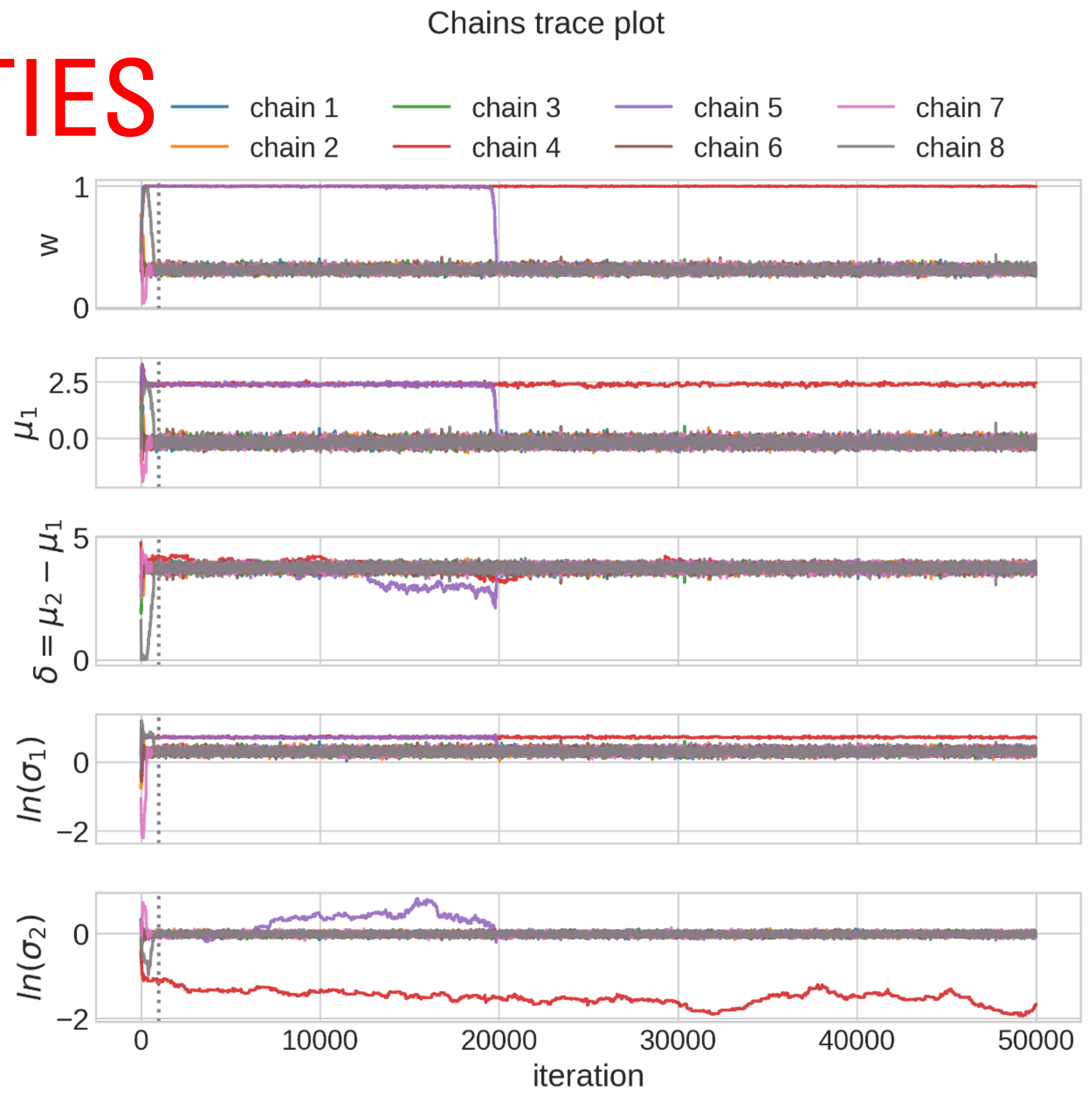
- $p(x|\boldsymbol{\theta}) = \sum_{\mathrm{i}} w_i \mathcal{N}\left(x \middle| \mu_{\mathrm{i}}, \sigma_{\mathrm{i}}^2\right)$

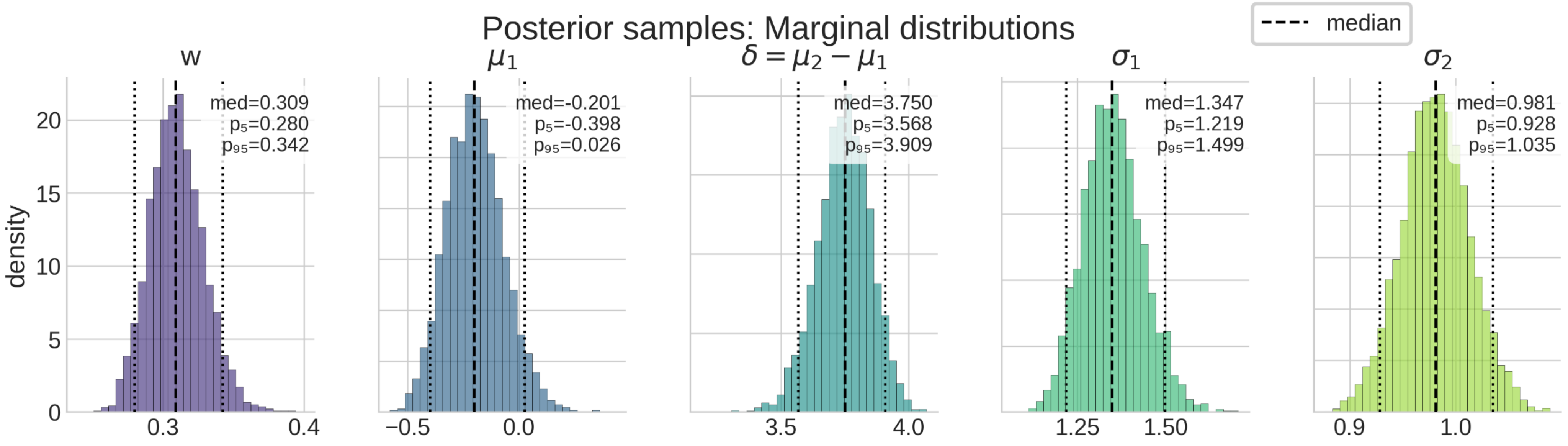
$$
\begin{aligned}
p(\,x_{obs} \mid \boldsymbol{\theta}, \sigma_{obs}\,) \quad &= \quad \int_{-\infty}^{+\infty} \mathcal{N}(\,x_{obs} \mid x, \sigma_{obs}^2\,) \sum_i w_i \; \mathcal{N}(\,x \mid \mu_i, \sigma_i^2\,)\, dx \\
&= \quad \sum_i w_i \int_{-\infty}^{+\infty} \mathcal{N}(\,x_{obs} \mid x, \sigma_{obs}^2\,)\; \mathcal{N}(\,x \mid \mu_i, \sigma_i^2\,)\, dx \\
&= \quad \sum_i w_i \; \mathcal{N}(\,x_{obs} \mid \mu_i, \sigma_i^2 + \sigma_{obs}^2\,)
\end{aligned}
$$

- The inference is analogous to Problem 1a; the values $\sigma_{\mathrm{obs},i}^2$ for $i = 1, \dots, N$ are known, so for each measurement we add $\sigma_{\mathrm{obs},i}^2$ to the corresponding variance.

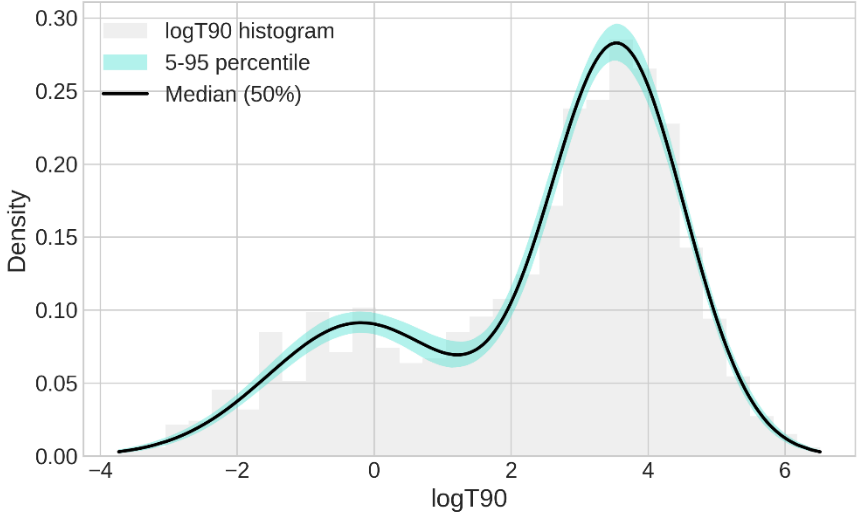
ENSEMBLE - UNCERTAINTIES

- Initial state validated on simulated data; real data stability uncertain.
- Ensemble of **8 chains** with **different initial states**.
- Initial states obtained by **Gaussian perturbation** (mean = base init; std = 10% of parameter range; clipped to prior boundaries).
- 2/8 chains stuck at $w \approx 1 \rightarrow$ discarded.
- **Thinning**: first lag where autocorrelation < 0 (max across parameters).
- **Burnin** = 1000.





Mean acceptace rate = 0.268
#samples = 4349



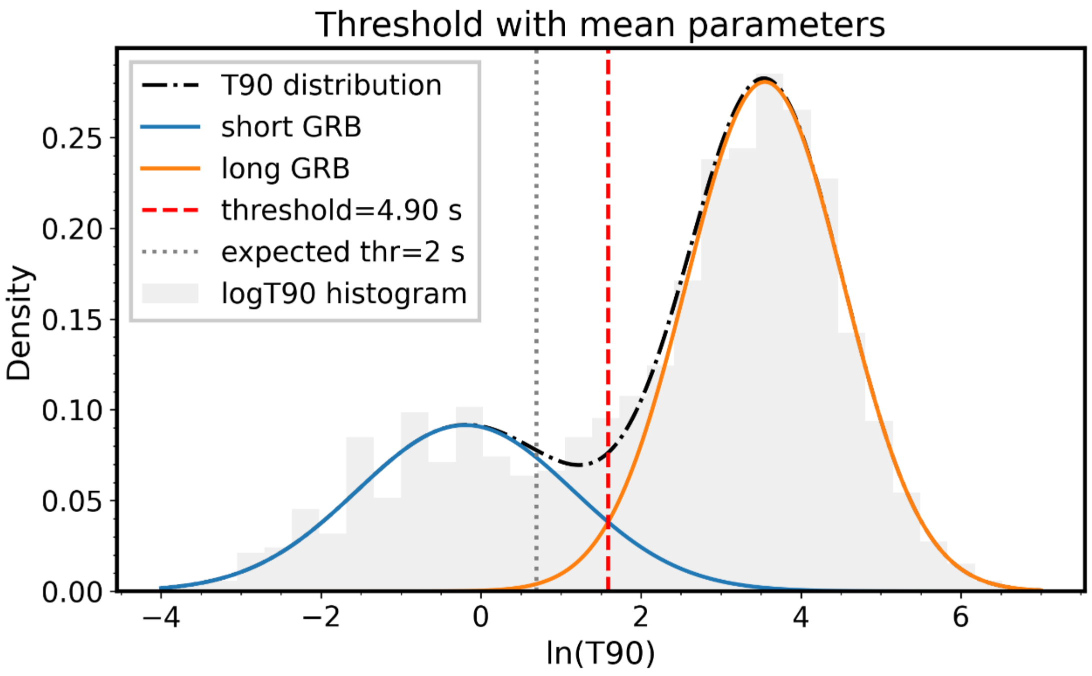
PARAMETER VALUES - UNCERTAINTIES

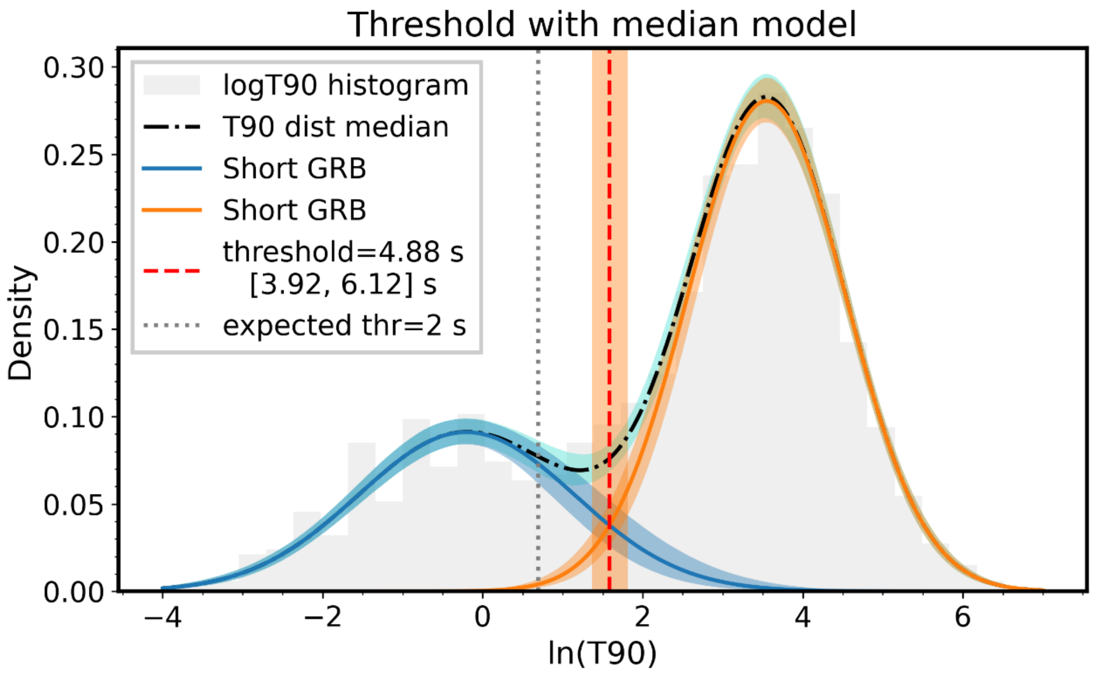
Parameters	Mean ± std	Median$\pm[p_{16}, p_{84}]$(68%)	$[p_5, p_{95}]$ (90%)
w_1	0.310 ± 0.019	$0.309^{+0.019}_{-0.018}$	[0.280, 0.342]
μ_1 [log s]	-0.20 ± 0.13	$-0.20^{+0.13}_{-0.12}$	[-0.40, 0.03]
δ_{12} [log s]	3.75 ± 0.10	$3.75^{+0.10}_{-0.10}$	[3.57, 3.91]
μ_2 [log s]	3.549 ± 0.045	$3.550^{+0.044}_{-0.045}$	[3.476, 3.623]
σ_1 [log s]	1.351 ± 0.085	$1.347^{+0.088}_{-0.080}$	[1.219, 1.499]
σ_2 [log s]	0.981 ± 0.032	$0.981^{+0.031}_{-0.031}$	[0.928, 1.035]

- Parameter estimates obtained when neglecting the uncertainty in x are **compatible** with those obtained after including that uncertainty.

PROBLEM 2 AND 3 - CLASSIFICATION

- **Problem 2:** compute the probability of GRB170817A (T_{90} = 2.0 s) of being a short GRB or a long GRB.
- **Problem 3:** determine the threshold value for *T90* to discriminate between short and long GRBs.





SHORT GRB PROBABILITY DISTRIBUTION

- $p(short|x,\boldsymbol{\theta}) = \frac{p(short|\boldsymbol{\theta})p(x|short,\boldsymbol{\theta})}{p(x|\boldsymbol{\theta})} = \frac{w_1\mathcal{N}\left(x\middle|\mu_1,\sigma_1^2\right)}{w_1\mathcal{N}\left(x\middle|\mu_1,\sigma_1^2\right)+(1-w_1)\mathcal{N}\left(x\middle|\mu_2,\sigma_2^2\right)}$

- $x_{obs} = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}\left(0,\sigma_{obs}^2\right)$

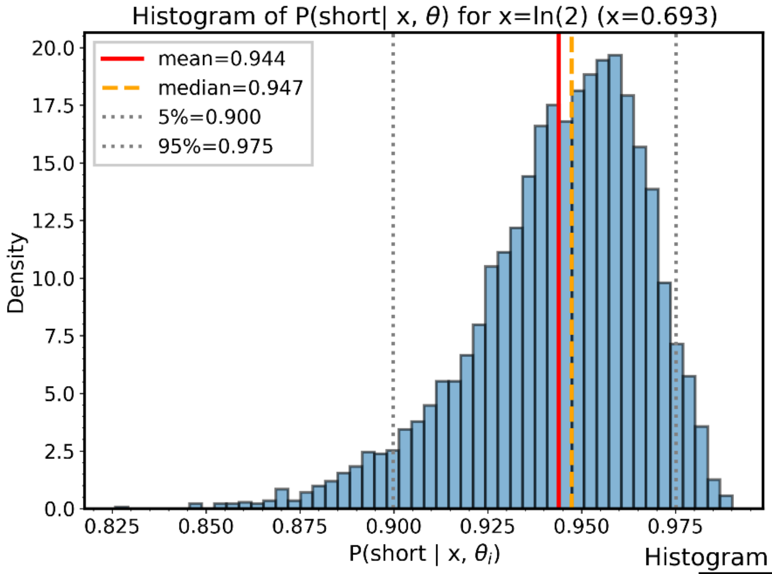
$$\implies \quad p(short|x_{obs},\boldsymbol{\theta}) = \frac{w_1\mathcal{N}\left(x_{obs}\middle|\mu_1,\sigma_1^2+\sigma_{obs}^2\right)}{w_1\mathcal{N}\left(x_{obs}\middle|\mu_1,\sigma_1^2+\sigma_{obs}^2\right)+(1-w_1)\mathcal{N}\left(x_{obs}\middle|\mu_2,\sigma_2^2+\sigma_{obs}^2\right)}$$

- The quantity is defined if the true parameters $\boldsymbol{\theta}$ are known, but in practice we only have posterior samples $\{\boldsymbol{\theta}_j\}_{\mathrm{j=1}}^{\mathrm{M}}$ from Metropolis.

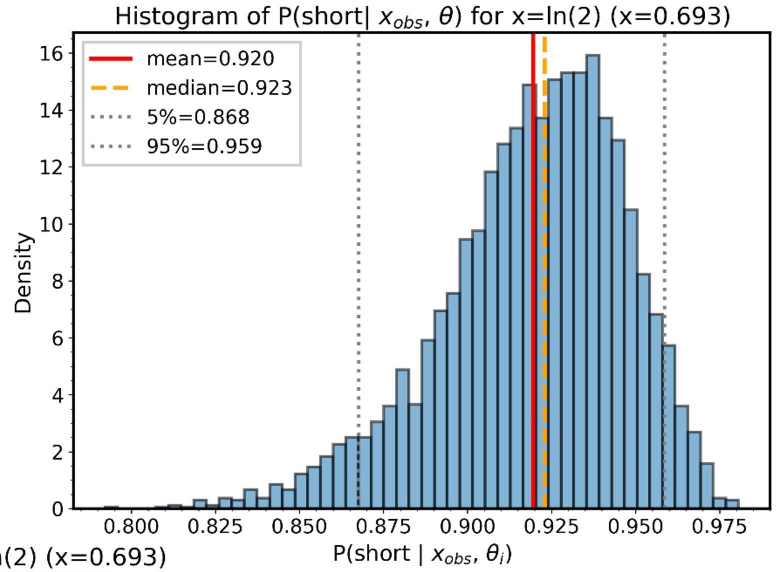
- Let the data be $\boldsymbol{D} \stackrel{\text{def}}{=} \{logT_{90_i}, \mathrm{dlog}T_{90_i}\}_{i=1}^{N}$. The value x_{obs} may or may not be included in $\boldsymbol{D}$.

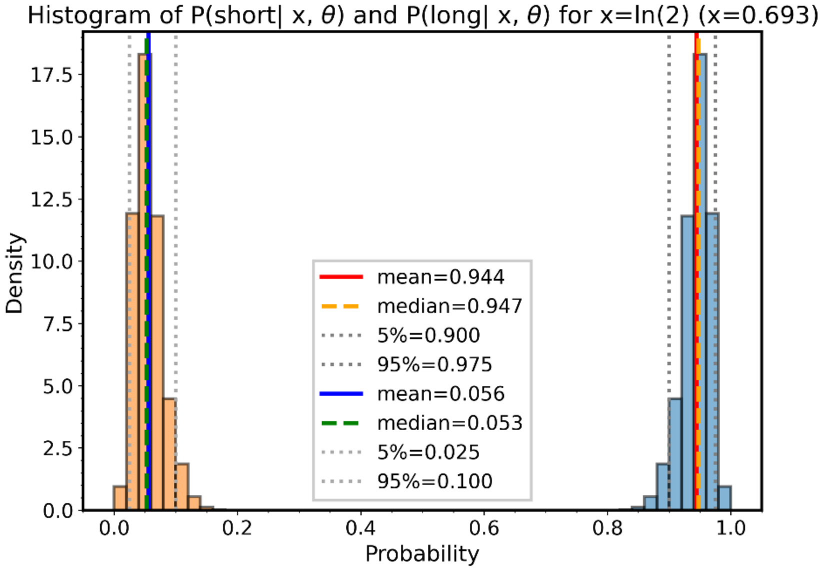
- $p(short|x_{obs},\boldsymbol{D}) = \int p(short|x_{obs},\boldsymbol{\theta})p(\boldsymbol{\theta}|\boldsymbol{D})d\boldsymbol{\theta} \approx \frac{1}{M}\sum_{j=1}^{M} p\left(short\middle|x_{obs},\boldsymbol{\theta_j}\right)$ $\qquad M = \#$ samples

- $p(long|x_{obs},\boldsymbol{D}) = 1 - p(short|x_{obs},\boldsymbol{D})$



Distributions of GRB170817A



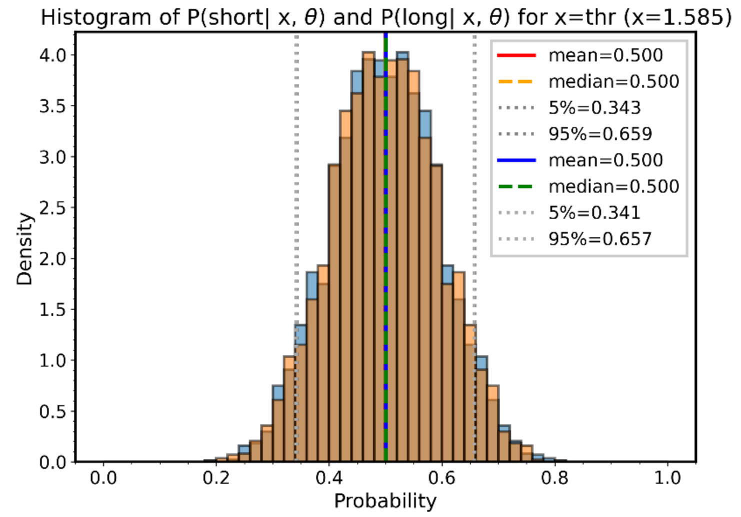


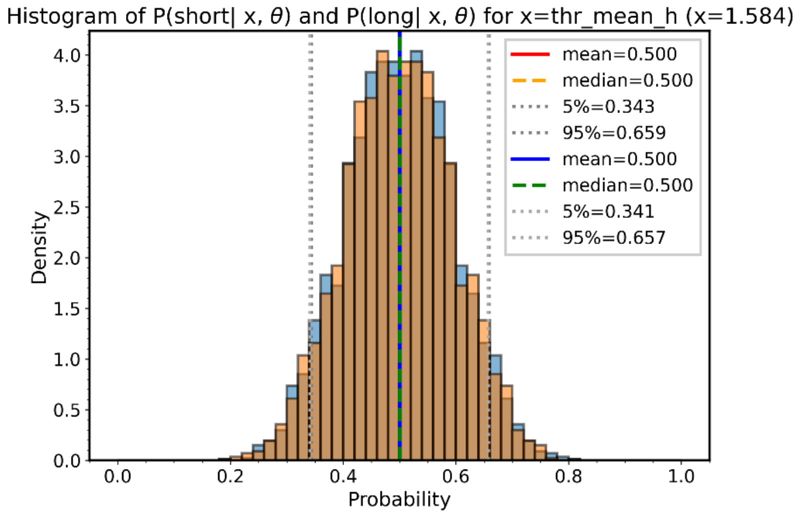
ESTIMATE THRESHOLD

- The **graphically estimated threshold** (4.88 s) was checked and found to match expectations: the posterior mean is 0.5 and the two class distributions overlap.
- **Probability-based threshold estimation - Method 1:**
  1. Create a grid of 10,000 points over the interval $x \in [-4,7]$.
  2. For each grid point x compute $p\left(short\middle|\mathrm{x}, \boldsymbol{\theta}_\mathrm{j}\right)$ and extract the postirior mean $\bar{\mathrm{h}}(x) \approx p(short|x, \boldsymbol{D})$.
  3. Extract the threshold using the **crossing method:**
     - Find two consecutive grid points x_1 and x_2 (with $x_2 \geq x_1$) such that $\bar{\mathrm{h}}(x)$ - 0.5 changes sign between them.
     - Estimate the threshold by **linear interpolation**:

$$\mathrm{thr\_mean\_h} = x_1 + (0.5 - \bar{\mathrm{h}}(x_1)) \frac{x_2 - x_1}{\bar{\mathrm{h}}(x_2) - \bar{\mathrm{h}}(x_1)}$$

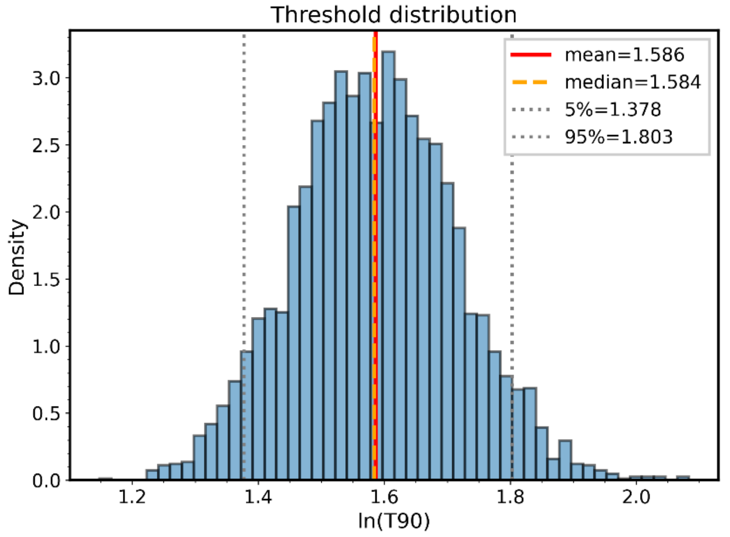
  4. This yields thr_mean_h $\simeq$ 4.87 s

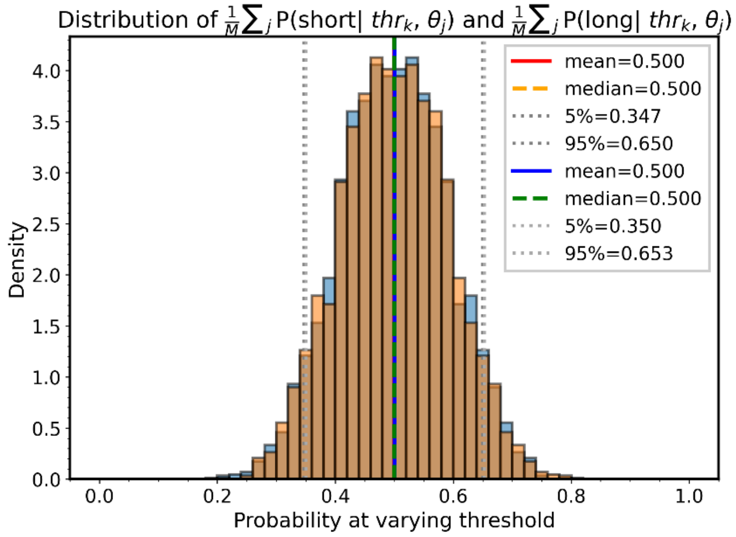




THRESHOLD DISTRIBUTION

- **Probability-based threshold estimation - Method 2:**
  1. Create a grid of 10,000 points over the interval $x \in [-4,7]$.
  2. For each grid point x compute $h_j(x) = p(short|\mathrm{x}, \boldsymbol{\theta}_\mathrm{j})$.
  3. For each sample j find the threshold using the **crossing method** (the x where $h_j(x) = 0.5$).
  4. This gives M thresholds $\{thr_j\}_{\mathrm{j=1}}^{\mathrm{M}}$; they form the **threshold distribution**. The result is: $thr = 4.87^{+0.57}_{-0.66}$ [3.96, 6.07] s.
- To check whether the threshold distribution matches expectations: for each thr_k compute $p(short|thr_k, \boldsymbol{\theta}_\mathrm{j})$ and extract the mean over the samples. Use the collection $\{\frac{1}{\mathrm{M}}\sum_\mathrm{j} p(short|thr_k, \boldsymbol{\theta}_\mathrm{j})\}_{\mathrm{k=1}}^{\mathrm{M}}$ as the **distribution of means**.





PROBLEM 4 – MODEL SELECTION

- **Problem 4:** Some authors propose a third class of GRBs, *intermediate.* Which of the two hypothesis is favored according to the available data?
- **H_0:** Mixture with **two** components $\implies$ $\boldsymbol{\theta}_0 = (w, \mu_1, \delta, \sigma_1, \sigma_2)$
- **H_1:** Mixture with **three** components $\implies$ $\boldsymbol{\theta}_1 = (\mathrm{v}_1, \mathrm{v}_2, \mu_1, \delta_{12}, \delta_{23}, \sigma_1, \sigma_2, \sigma_3)$
- $\frac{p(H_1|\boldsymbol{D})}{p(H_0|\boldsymbol{D})} = \frac{p(\boldsymbol{D}|H_1)p(H_1)}{p(\boldsymbol{D}|H_0)p(H_0)}$ ***Odds ratio***
- Assuming equal prior probability for the models: $p(H_1) = p(H_0)$

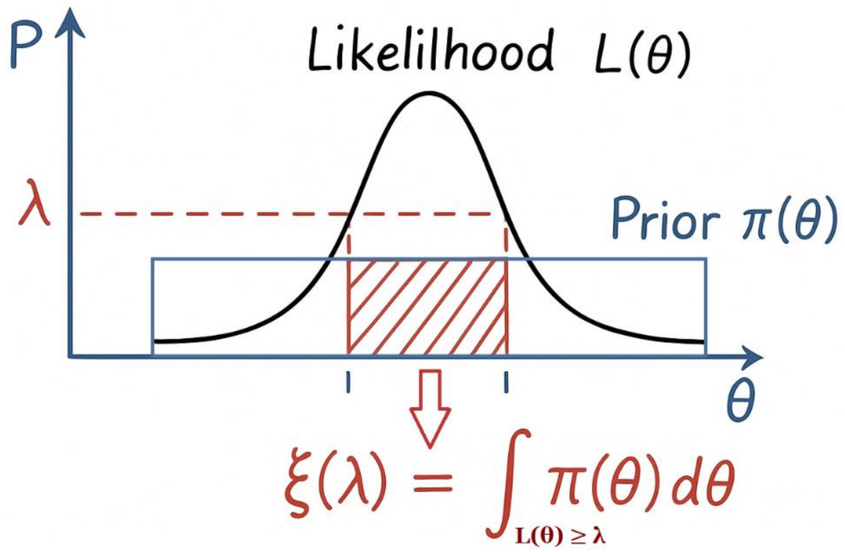
$$\implies \quad \frac{p(H_1|\boldsymbol{D})}{p(H_0|\boldsymbol{D})} = \frac{p(\boldsymbol{D}|H_1)}{p(\boldsymbol{D}|H_0)} = \frac{\int p(\boldsymbol{D}|H_1, \boldsymbol{\theta_1})p(\boldsymbol{\theta_1}|H_1)d\boldsymbol{\theta_1}}{\int p(\boldsymbol{D}|H_0, \boldsymbol{\theta_0})p(\boldsymbol{\theta_0}|H_0)d\boldsymbol{\theta_0}} = \frac{Z_1}{Z_0} \quad \textbf{\textit{Bayes Factor}}$$

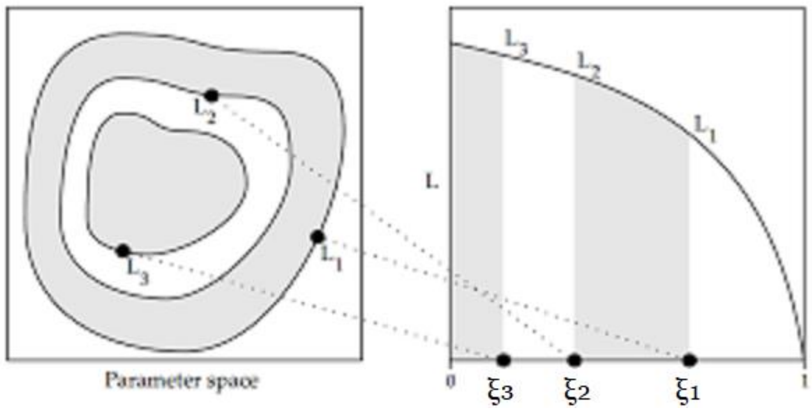
- **Occam's razor:** even if H_1 has higher likelihood (a more complex model can fit the data better), its prior can reduce the model evidence Z_1, because $p(\boldsymbol{\theta_1}|H_1) \propto 1/V_1$ and $V_1 \gg V_0$, where V_i is the parameter space volume.

NESTED SAMPLING (1/2)

- $Z = \int L(\theta)\pi(\theta)d\theta = \int_0^1 L(\xi)d\xi$ where $d\xi = \pi(\theta)d\theta$
- $L(\xi)$ is **one-dimensional** and **monotonically decreasing**.

1. Draw N_L live points from the prior and set $\log\xi_0 = 0$.
2. Compute the likelihood for each live point; sort in decrising order and pick the **lowest value** $\log L_{min,\mathrm{i}}$.
3. Discart the point with $\log L_{min,\mathrm{i}}$ and **draw a new sample** from the prior constrained to $L > L_{min,i}$.
4. Compute $\log\xi_\mathrm{i}$:
   - $\{\xi\}^i \sim \mathcal{U}(0, \xi_{\mathrm{i}-1}) \quad \Rightarrow \quad \{t\}^i = \{\xi/\xi_{\mathrm{i}-1}\}^i \sim \mathcal{U}(0,1)$
   - $t_i = max\{t\}^i \quad \Rightarrow \quad t_i \sim Beta(N_L, 1) = N_L t^{N_L - 1}$
   - $\mathbb{E}[log t_i] = N_L \int_0^1 t^{N_L-1} log t \, dt = -\frac{1}{N_L} \quad \Rightarrow \quad \log\xi_\mathrm{i} = \log\xi_{\mathrm{i}-1} - \frac{1}{N_L}$





NESTED SAMPLING (2/2)

5. Compute **weights**: $w_i = \xi_{i-1} - \xi_i \quad \Rightarrow \quad logw_i = log\left(e^{log\xi_{i-1}} - e^{log\xi_i}\right)$

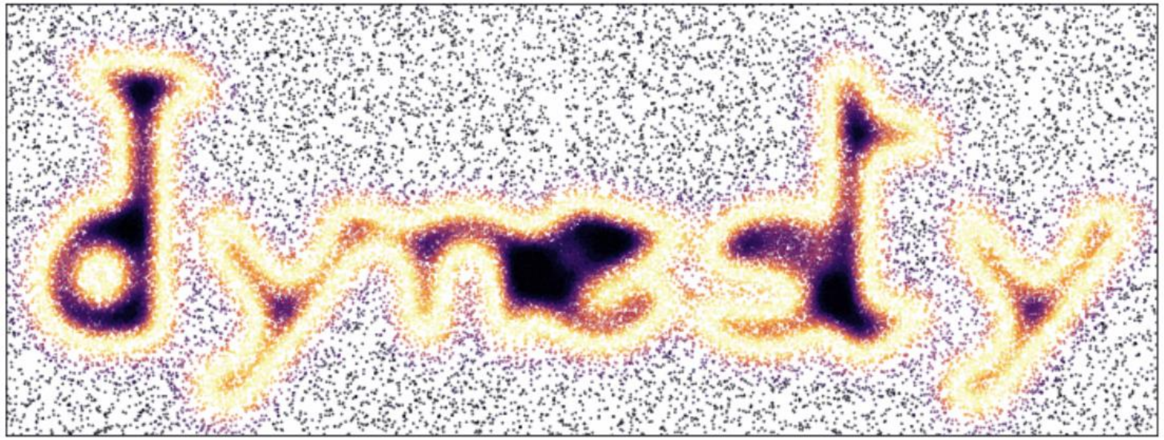
6. Accumulate **evidence**: $Z_i = Z_{i-1} + w_i L_{min,i} \quad \Rightarrow \quad logZ_i = log\left(e^{logZ_{i-1}} + e^{logw_i + logL_{min,i}}\right)$

7. **Stopping condition**: $\Delta logZ_i \stackrel{\text{def}}{=} log(Z_i + \Delta Z_i) - logZ_i \lesssim log\left(Z_i + L_{max,i}\xi_i\right) - logZ_i < \varepsilon$
   - ε is the tolerance and $\Delta Z_i \lesssim L_{max,i}\xi_i$ is the **estimated remaining evidence**, where $L_{max,i}$ is the maximum value of the likelihood among the live points at iteration i and ξ_i is the remaning prior volume.
   - The condition $\Delta Z_i \lesssim L_{max,i}\xi_i$ is an **upper bound** and is valid only if there is **not** an **extremely narrow likelihood peak** within the remaining prior volume that has not yet been discovered by the N_L live points.

8. Residual: $Z = Z_\mathrm{K} + \xi_\mathrm{K}\bar{L} \quad \Rightarrow \quad logZ = log\left(e^{logZ_\mathrm{K}} + e^{log\xi_\mathrm{K} + log\bar{L}}\right)$
   - K indicates the final iteration and $\bar{L}$ is the mean likelihood of the remaining live points.

9. **Posterior samples**: $p(\boldsymbol{\theta}_i) = w_i L_{min,i}/Z \quad \Rightarrow \quad p(\boldsymbol{\theta}_i) = e^{logw_i + logL_{min,i} - logZ}$
   - From the available **weighted samples** ($K + N_L$ = #iterations + residuals), draw M **unweighted posterior samples** using `numpy.random.choice` with probabilities $p = \{p(\boldsymbol{\theta}_i)\}$.

LIBRARY: DYNESTY

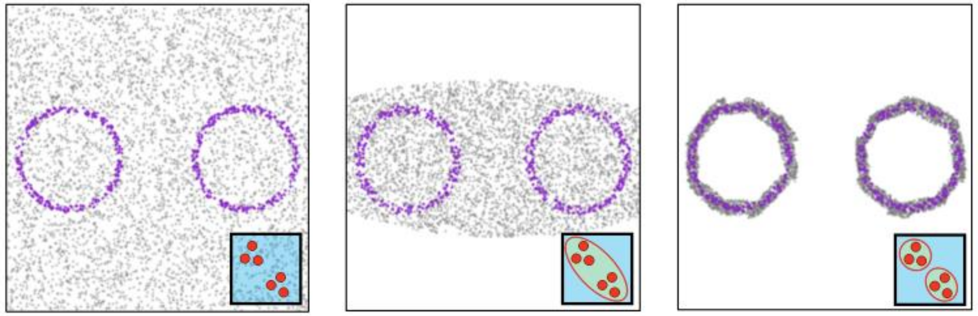


- Class: `NestedSampler`
- #Live points $\Rightarrow N_L = 1200$
- Tolerance $\Rightarrow \varepsilon = 10^{-2}$
- Bound $\Rightarrow$ Multiple ellipsoids (‘multi’)
- Sample $\Rightarrow$ Random walk (‘rwalk’)

PRIOR

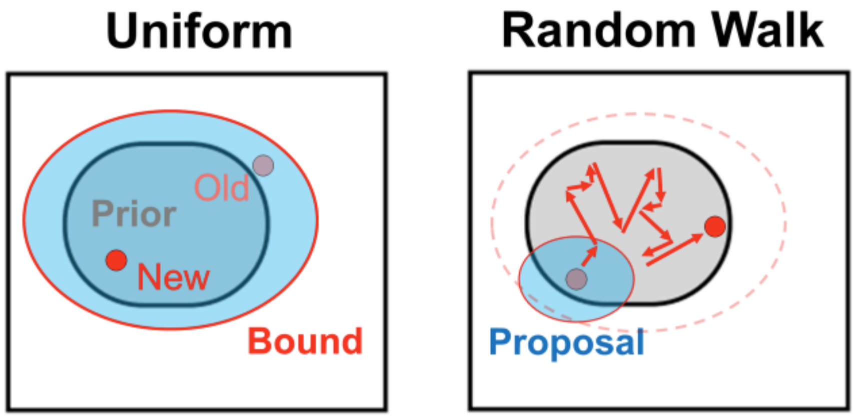
- `dynesty` uniformly draws prior samples from a multidimensional unit cube and then applies a transformation using a user-defined function.
- For model selection, the priors must be **normalized**.
- Weights are constructed using a **strick breaking process:**
  - $w_1 = v_1, \quad w_2 = (1 - v_1)v_2, \quad w_3 = 1 - w_1 - w_2$
  - $p(v_{\mathrm{i}}) = Beta(2,2) = 6v_i(1 - v_i)$
- $\mathrm{p}(\mu_1) \; = \mathcal{U}(min(logT_{90}), \; max(log\mathrm{T}_{90}))$
- $\mathrm{p}(\delta_{12}|\mu_1) = \mathcal{U}(0, max(log\mathrm{T}_{90}) - \mu_1)$
- $\mathrm{p}(\delta_{23}|\delta_{12}, \mu_1) = \mathcal{U}(0, max(log\mathrm{T}_{90}) - (\mu_1 + \delta_{12}))$
- $p(log\sigma_i) = \mathcal{U}\big(log(0.1), log(6)\big)$
- For H_0 the priors are the same, but $w_1 = w$ and $w_2 = 1 - w$.

BOUND



- **Bound = 'none'**
  - Inefficient: the rejection probability grows whit $L_{min,i}$.
- **Bound = 'single'**
  - Create an ellipsoid using mean and covariance of the live points.
  - Works well for approximately Gaussian, unimodal posteriors; inefficient for multimodal posteriors.
- **Bound = 'multi'**
  - Build **multiple ellipsoids** by recursive clustering:
    1. Compute the mean and covariance of the current live points and construct an ellipsoid.
    2. Select the two points corresponding to the opposite ends of the major axis of the ellipsoid.
    3. Apply k-means using those two points (k=2) and assign them to the two clusters.
    4. Construct an ellipsoid for each cluster.
    5. If the decrease in volume exceeds a threshold, repeat from step 2 on each ellipsoid (further subdivide); otherwise stop.

SAMPLE



- **Sample = ‘unif’**
  - **Rejection sampling** inside the bound.
- **Sample = ‘rwalk’**
  - The **proposal** is a uniform ellipsoid centered on the point to be replaced and with the same size as the bound. If *bound* = *‘multi’*, randomly pick one of the ellipsoids that intersect the point.
  - The **number of chain steps** per iteration is set by the parameter *walks* (default *walks* = 25).
  - The **bound is updated** after $round(0.15 \times walks) \times N_L$ likelihood calls:
    - $N_L = 1200, \quad walks = 25 \quad \Rightarrow \quad round(0.15 \times walks) \times N_L = 4500$ likelihood calls (180 iterations)
  - The **proposal scale** is adapted each iteration:

$$s_{i+1} = s_i \times exp\left(\frac{ratio - target}{d \times target}\right) \quad \Rightarrow \quad log(s_{i+1}) = log(s_i) + \eta(ratio - target),$$

where $\eta = 1/(d \times target)$, $ratio$ is the acceptance ratio in that iteration, $target$ (default is 0.5) is the desired acceptance ratio, and d is the parameter dimension.

RESULTS

- The evidences Z_i were computed **5 times** for each hypothesis.
- Validation on **simulated data**:
  - Data generated from **2 components**:

    $$\text{logBF} = -2.63 \pm 0.19 < 0 \quad \Rightarrow \quad p(H_0|\boldsymbol{D}) > p(H_1|\boldsymbol{D})$$

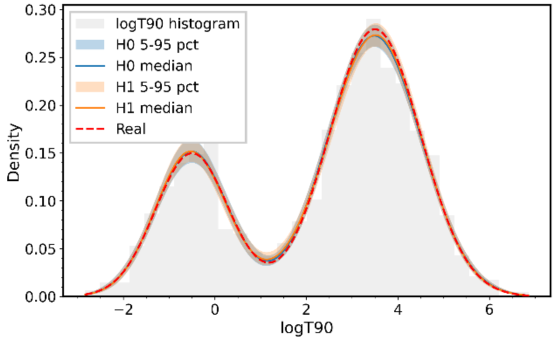
  - Data generated from **3 components**:

    $$\text{logBF} = 6.07 \pm 0.18 > 0 \quad \Rightarrow \quad p(H_0|\boldsymbol{D}) < p(H_1|\boldsymbol{D})$$

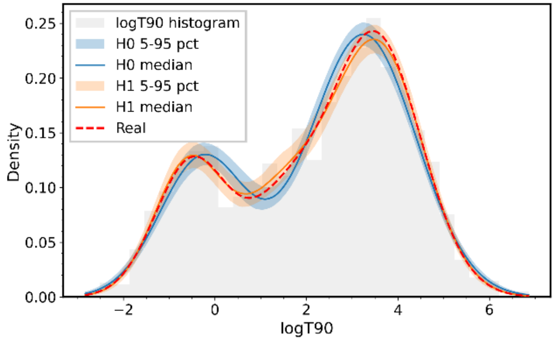
- On real data:

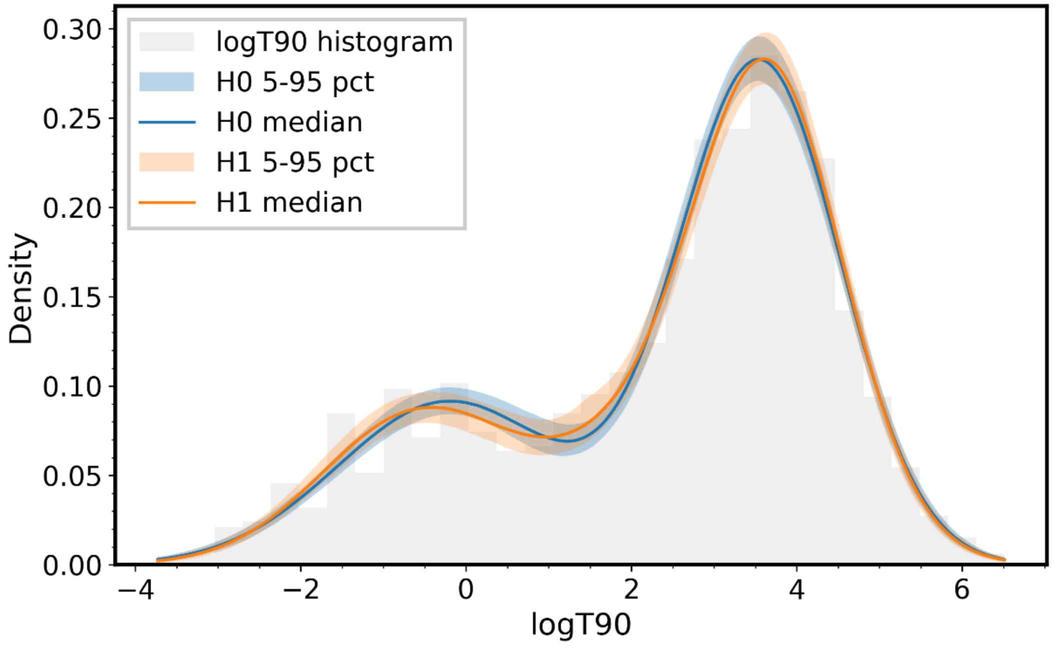
  $$\text{logBF} = -1.75 \pm 0.12 < 0 \quad \Rightarrow \quad p(H_0|\boldsymbol{D}) > p(H_1|\boldsymbol{D})$$

Data generated from **2 components**



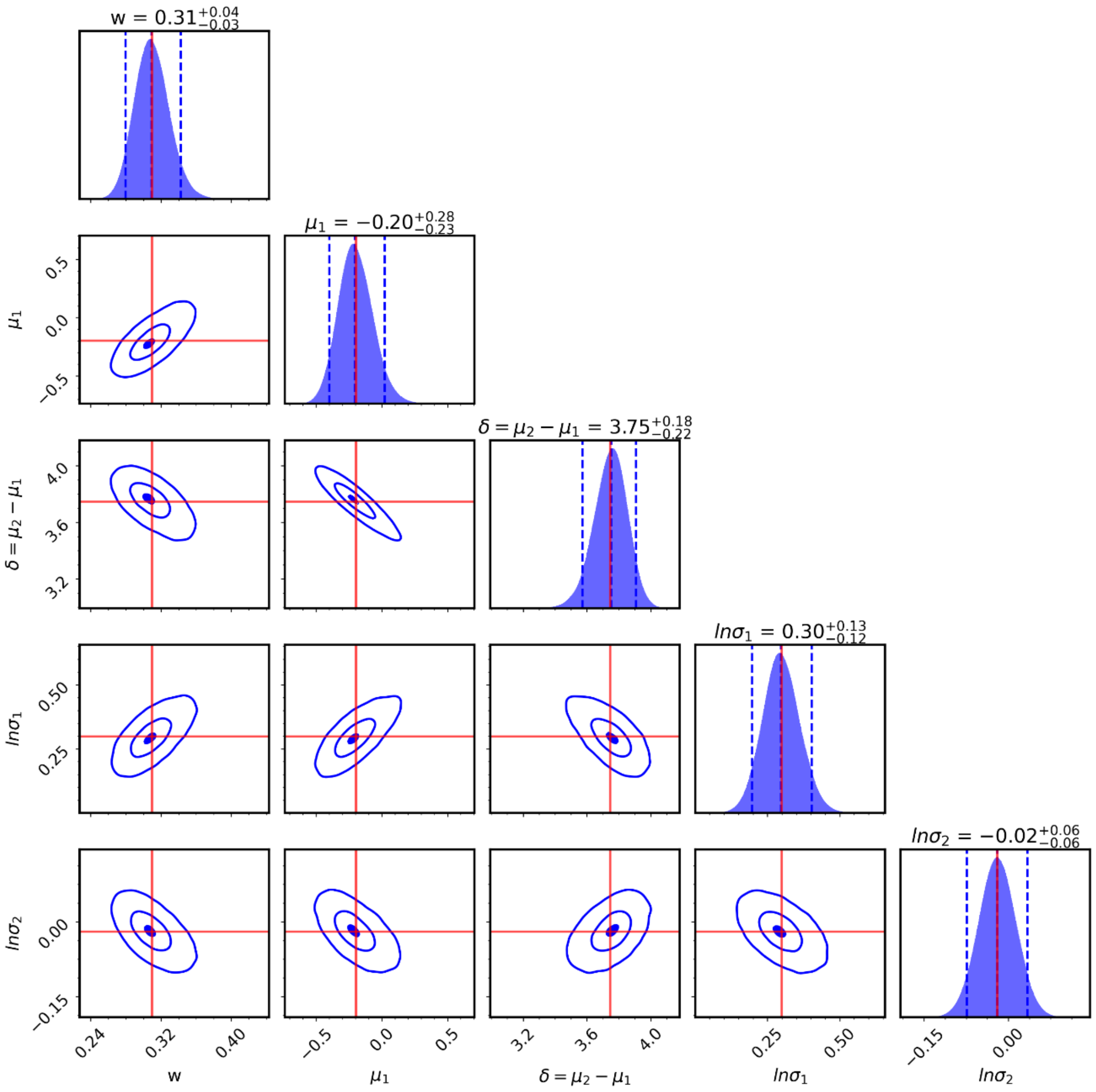
Data generated from **3 components**



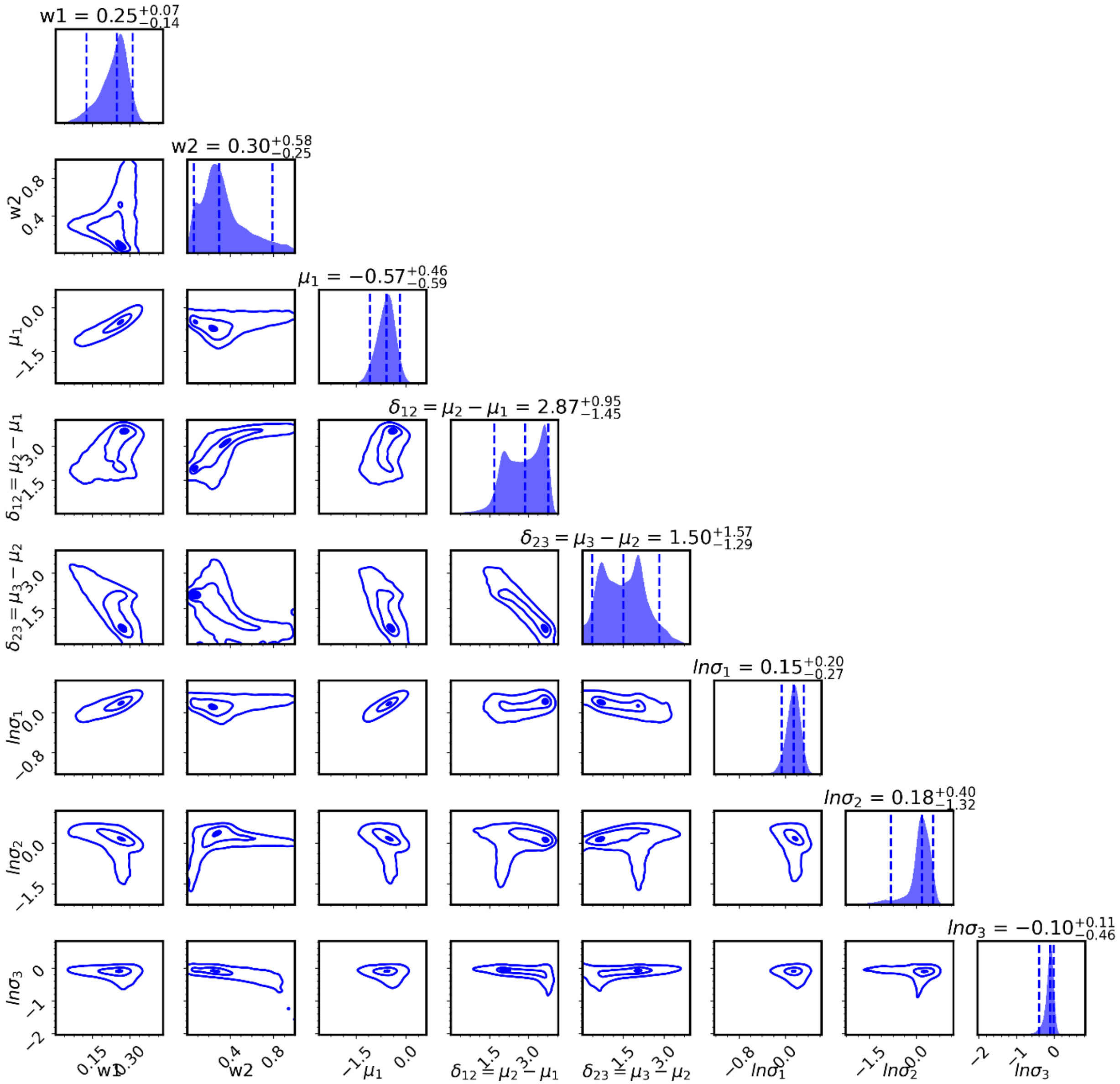


Corner plot - Model H_0

(Metropolis posterior means in red)



Corner plot – Model H_1



PROBLEM 5 - CLUSTERING

- **Problem 5:** Another possibility is to classify GRBs in **soft** and **hard**, according to the hardness ratio:

$$HR = \frac{F_{50-100\ \mathrm{keV}}}{F_{20-50\ \mathrm{keV}}}$$

where F is the measured flux in a certain energy interval. Study the bimodal distribution in the *log(T90) - log(HR)* space.

GIBBS SAMPLING

- A **Markov Chain Monte Carlo (MCMC)** method to draw samples from a **joint posterior** by iteratively sampling each variable from its **conditional distribution** given the current values of the others.
- **Assignments**: $\boldsymbol{z} = (z_1, \dots, z_{\mathrm{N}}),\ z_i \in \{1, \dots, K\}$ $\quad K =$ #Gaussian components
- **Parameters**: weights $\boldsymbol{w} = (w_1, \dots, w_K)$, gaussian parameters $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\sigma}^2) = (\{\mu_k\}_{k=1}^K, \{\sigma_k^2\}_{k=1}^K)$
- Principle:

$$p(\gamma_j | \gamma_{\neg j}, \boldsymbol{x}) = p(\gamma_j, \gamma_{\neg j} | \boldsymbol{x}) / p(\gamma_{\neg j} | \boldsymbol{x}) \propto p(\gamma_j, \gamma_{\neg j} | \boldsymbol{x}) = p(\boldsymbol{\gamma} | \boldsymbol{x})$$

$$p(\boldsymbol{\gamma}|\boldsymbol{x}) = p(\boldsymbol{\theta}, \boldsymbol{w}, \boldsymbol{z}|\boldsymbol{x}) \propto p(\boldsymbol{x}|\boldsymbol{\theta}, \boldsymbol{w}, \boldsymbol{z}) p(\boldsymbol{z}|\boldsymbol{w}) p(\boldsymbol{w}) p(\boldsymbol{\theta}) =$$

$$= \left(\prod_{i=1}^{N} p(x_i | z_i, \boldsymbol{\theta})\right)\left(\prod_{i=1}^{N} w_{z_i}\right) p(\boldsymbol{w}) p(\boldsymbol{\theta}) = p(\boldsymbol{\theta}) p(\boldsymbol{w}) \prod_{i=1}^{N} w_{z_i} \mathcal{N}(x_i | \mu_{z_i}, \sigma_{z_i}^2)$$

- For each γ_j compute the posterior discarting all the pieces that depends only on $\gamma_{\neg j}$.

GIBBS SAMPLING ALGORITHM

1. Initialize $\boldsymbol{\theta}^{(0)}$, $\boldsymbol{w}^{(0)}$ and $\boldsymbol{z}^{(0)}$.
2. For iterations $t = 1,2, \dots, M$:
   - For each i: draw $z_i^{(t)} \sim p\left(z_i \middle| x_i, \boldsymbol{w}^{(t-1)}, \boldsymbol{\theta}^{(t-1)}\right)$.
   - Draw $\boldsymbol{w}^{(t)} \sim p\left(\boldsymbol{w} \middle| \boldsymbol{z}^{(t)}\right)$.
   - For each k: draw $\left(\mu_k^{(t)}, \sigma_k^{2(t)}\right) \sim p\left(\mu_k, \sigma_k^2 \middle| \boldsymbol{x}_k\right)$ where $\boldsymbol{x}_k = \{x_i \colon z_i^{(t)} = k\}$
3. Remove burn-in and apply thinning.

CONDITIONAL PROBABILITIES (1/2)

1. For each k: $p(z_i = \mathrm{k}|x_i, \boldsymbol{w}, \boldsymbol{\theta}) = \dfrac{w_\mathrm{k}\mathcal{N}\left(x_i\middle|\mu_\mathrm{k}, \sigma_\mathrm{k}^2\right)}{\sum_{h=1}^{K} w_h\mathcal{N}\left(x_i\middle|\mu_\mathrm{h}, \sigma_\mathrm{h}^2\right)}$
   - Draw one component with `numpy.random.choice` using probabilities $p = \{p(z_i = \mathrm{k}|x_i, \boldsymbol{w}, \boldsymbol{\theta})\}_{k=1}^{K}$.
2. $p(\boldsymbol{w}|\boldsymbol{z}) \propto p(\boldsymbol{z}|\boldsymbol{w})p(\boldsymbol{w}) = \left(\prod_{i=1}^{N} w_{z_i}\right) Dir(\boldsymbol{w}|\alpha/K) \propto \left(\prod_{\mathrm{k}=1}^{\mathrm{K}} w_k^{n_k}\, w_k^{\alpha/\mathrm{K}\,-\,1}\right) \propto Dir(\boldsymbol{w}|\alpha/K + n_k)$
   - n_k is the number of points assigned to component k.
   - The prior is a **Dirichlet distribution** with hyperparameter α/K for all components: $p(\boldsymbol{w}) \rightarrow p(\boldsymbol{w}|\alpha)$.
3. Prior: $\mathrm{p}(\theta_k) = p(\mu_k)\mathrm{p}(\sigma_k^2) \propto 1/\sigma_k^2$ where $p(\mu_k) \propto 1$, $p\left(log(\sigma_k^2)\right) \propto 1 \Rightarrow \mathrm{p}(\sigma_k^2) \propto 1/\sigma_k^2$

- $p\left(\mu_k, \sigma_k^2\middle|\boldsymbol{x}_k\right) \propto p(\mu_k)p\left(\sigma_k^2\right)p\left(\boldsymbol{x}_k\middle|\mu_k, \sigma_k^2\right) =$

$$= \frac{1}{\sigma_k^2}\prod_{i=1}^{n_k}\frac{1}{\sqrt{2\pi\sigma_k^2}}exp\left(-\frac{1}{2}\frac{\left(x_{k,i} - \mu_k\right)^2}{\sigma_k^2}\right) \propto (\sigma^2)^{-\left(\frac{n_k}{2}+1\right)}exp\left(-\frac{1}{2\sigma_k^2}\sum_{i=1}^{n_k}\left(x_{k,i} - \mu_k\right)^2\right)$$

CONDITIONAL PROBABILITIES (2/2)

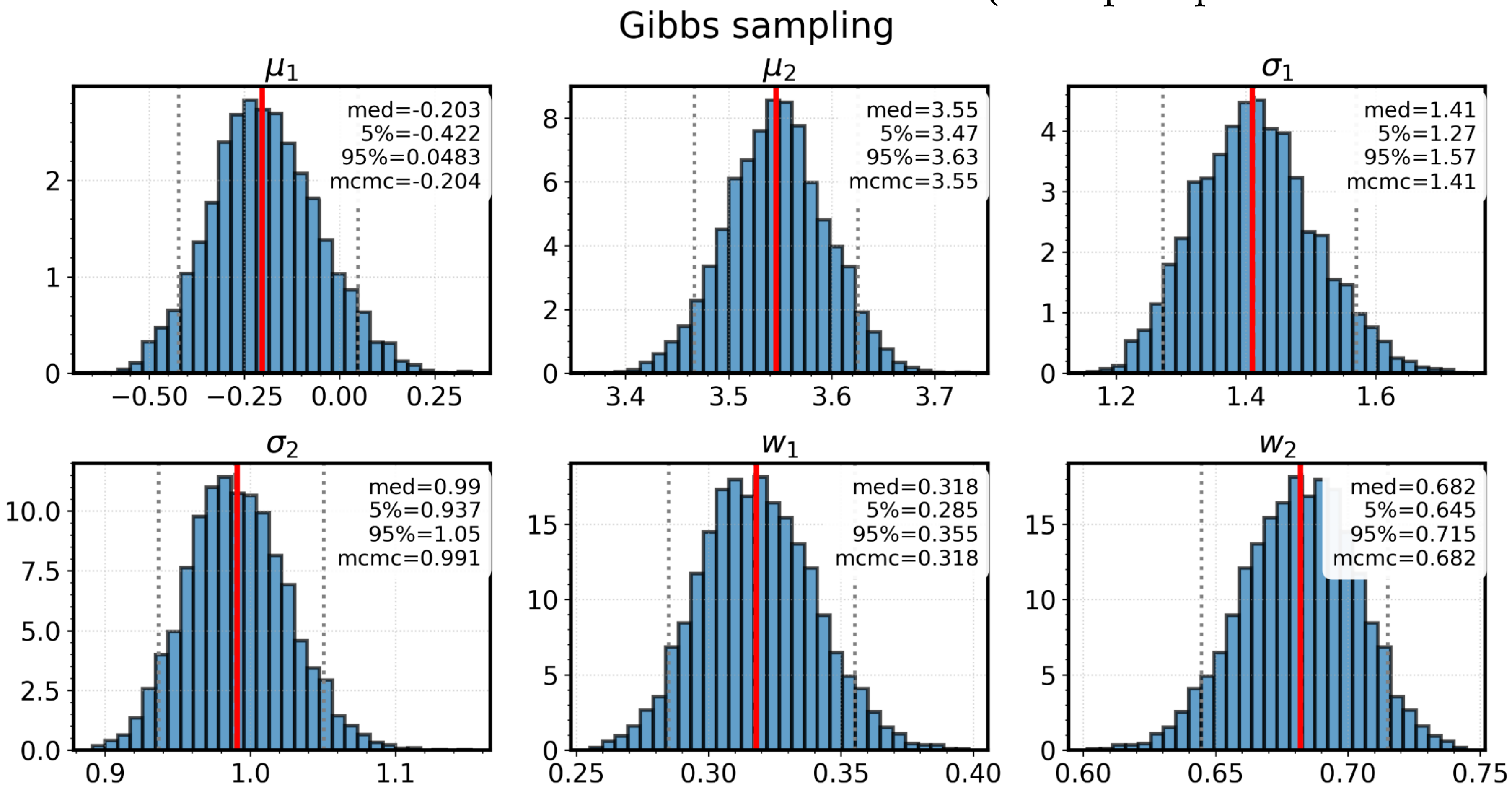
- Marginalize over μ_k to obtain a distribution only in σ_k^2.
- $$p(\sigma_k^2|\boldsymbol{x_k}) = \int_{-\infty}^{+\infty} p(\mu_k, \sigma_k^2|\boldsymbol{x}_k) d\mu_k \propto (\sigma^2)^{-\left(\frac{n_k}{2}+1\right)} e^{-\frac{S_k}{2\sigma_k^2}} \int_{-\infty}^{+\infty} exp\left(-\frac{n_k}{2\sigma_k^2}(\overline{x}_k - \mu_k)^2\right) d\mu_k \propto$$
  $$\propto (\sigma_k^2)^{-\left(\frac{n_k}{2}+1\right)} e^{-\frac{S_k}{2\sigma_k^2}} (\sigma_k^2)^{\frac{1}{2}} = (\sigma_k^2)^{-\left(\frac{n_k-1}{2}+1\right)} e^{-\frac{S_k}{2\sigma_k^2}} \propto \text{Inv} - \text{Gamma}(\sigma_k^2|\alpha_k = \frac{n_k-1}{2}, \beta_k = \frac{S_k}{2}).$$
  - The relation used is $\sum_i (x_{k,i} - \mu_k)^2 = S_k + n_k(\overline{x}_k - \mu_k)^2$, where $\overline{x}_k = (\sum_i x_{k,i})/n_k$ and $S_k = \sum_i (x_{k,i} - \overline{x}_k)^2$.
- Once σ_k^2 is sampled, sample μ_k from its conditional distribution:
  - $p(\mu_k|\sigma_k^2, \boldsymbol{x}_k) \propto exp\left(-\frac{n_k}{2\sigma_k^2}(\overline{x}_k - \mu_k)^2\right) \propto \mathcal{N}(\mu_k|\overline{x}_k, \sigma_k^2/n_k)$

$$\Rightarrow p(\mu_k, \sigma_k^2|\boldsymbol{x}_k) \propto p(\mu_k|\sigma_k^2, \boldsymbol{x}_k) p(\sigma_k^2|\boldsymbol{x}_k) = \mathcal{N}(\mu_k|\overline{x}_k, \sigma_k^2/n_k) \times \text{Inv} - \text{Gamma}\left(\sigma_k^2 \middle| \alpha_k = \frac{n_k - 1}{2}, \beta_k = \frac{S_k}{2}\right).$$

- This result requires $\alpha > 0 \Rightarrow n_k > 1$, otherwise Inv-Gamma normalization diverges.

RESULTS

(Metropolis posterior means in red)



COLLAPSED GIBBS SAMPLING

- If we are interested only in the assignment (clustering) of the data, we can marginalize out all model parameters and compute the conditional distribution of z given the data.

- $$\begin{aligned} p(z_i = k | z_{\neg i}, \boldsymbol{x}, \alpha) &= \frac{p(z_i = k, x_i | z_{\neg i}, \boldsymbol{x}_{\neg i}, \alpha)}{p(x_i | z_{\neg i}, \boldsymbol{x}_{\neg i}, \alpha)} \propto p(z_i = k, x_i | z_{\neg i}, \boldsymbol{x}_{\neg i}, \alpha) = \\ &= p(x_i | z_i = k, z_{\neg i}, \boldsymbol{x}_{\neg i}, \alpha) p(z_i = k | z_{\neg i}, \boldsymbol{x}_{\neg i}, \alpha) \end{aligned}$$

  - $p(z_i = k | z_{\neg i}, \boldsymbol{x}_{\neg i}, \alpha) = \int p(z_i = k | \boldsymbol{w}, \cancel{z_{\neg i}}, \cancel{\boldsymbol{x}_{\neg i}}, \cancel{\alpha}) p(\boldsymbol{w} | z_{\neg i}, \cancel{\boldsymbol{x}_{\neg i}}, \alpha) d\boldsymbol{w} = p(z_i = k | z_{\neg i}, \alpha)$

  - $$\begin{aligned} p(x_i | z_i = k, z_{\neg i}, \boldsymbol{x}_{\neg i}, \alpha) &= \iint p(x_i | \mu_k, \sigma_k^2, z_i = k, \cancel{z_{\neg i}}, \cancel{\boldsymbol{x}_{\neg i}}, \cancel{\alpha}) p(\mu_k, \sigma_k^2 | z_i = k, \cancel{z_{\neg i}}, \boldsymbol{x}_{\neg i}, \cancel{\alpha}) d\mu_k d\sigma_k^2 = \\ &= p(x_i | z_i = k, \boldsymbol{x}_{\neg i}) = p(x_i | \{x_j : ; z_j = k, j \neq i\}) \end{aligned}$$

$$\Rightarrow p(z_i = k | z_{\neg i}, \boldsymbol{x}, \alpha) \propto p(z_i = k | z_{\neg i}, \alpha) p(x_i | \{x_j : z_j = k, j \neq i\})$$

MARGINALIZE OVER THE WEIGHTS

- $p(z_i = k | z_{\neg i}, \alpha) \rightarrow$ Probability that point i belongs to cluster k given the assignments of all other points.
- The Dirichlet normalizing constant is: $B(\boldsymbol{\alpha}) = \int \prod_{j=1}^{K} w_j^{\alpha_j - 1} \, d\boldsymbol{w} = \frac{\prod_{j=1}^{K} \Gamma(\alpha_j)}{\Gamma\left(\sum_{j=1}^{K} \alpha_j\right)}$
- $p(\boldsymbol{z}|\alpha) = \int p(\boldsymbol{z}|\boldsymbol{w}) p(\boldsymbol{w}|\alpha) d\boldsymbol{w} = \frac{1}{B(\alpha/\mathrm{K})} \int \prod_{j=1}^{K} w_j^{\frac{\alpha}{\mathrm{K}} + n_j - 1} \, d\boldsymbol{w} = \frac{B(\alpha/\mathrm{K} + \boldsymbol{n})}{B(\alpha/\mathrm{K})},$ where $\boldsymbol{n} = (n_1, \dots, n_K)$
- $p(z_i = k | z_{\neg i}, \alpha) = \frac{p(\boldsymbol{Z}|\alpha)}{p(Z_{\neg i}|\alpha)} = \frac{B(\alpha/\mathrm{K} + \boldsymbol{n})}{B(\alpha/\mathrm{K} + \boldsymbol{n}_{\neg i})}$
  - where $n_{j\neg i}$ is the number of points in component j excluding point i (if point i is not in component j: $n_{j\neg i} = n_j$)

$$\Rightarrow p(z_i = k | z_{\neg i}, \alpha) = \frac{B(\alpha/\mathrm{K} + \boldsymbol{n})}{B(\alpha/\mathrm{K} + \boldsymbol{n}_{\neg i})} = \frac{\Gamma(\alpha/K + n_k)}{\Gamma(\alpha/K + n_{k\neg i})} = \frac{n_{k\neg i} + \alpha/K}{N - 1 + \alpha}$$

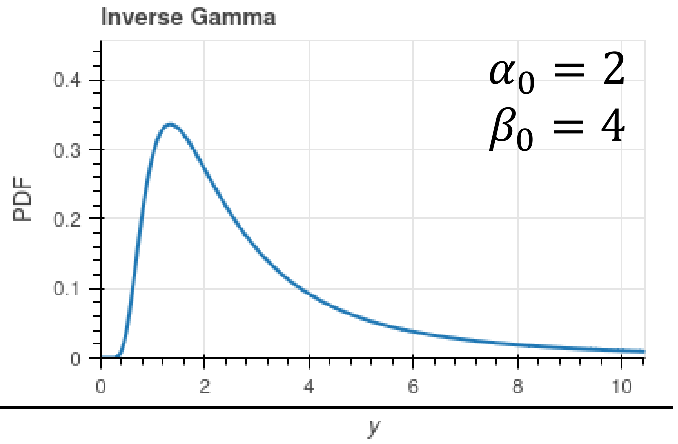
- The probability that $z_i = k$ is proportional to the number of points in components k ($n_{k\neg i}$).

MARGINALIZE OVER μ_k AND σ_k^2 (1/3)

- $p\left(x_i \middle| \{x_j\colon z_j = k, j \neq i\}\right) = p(x_i|\boldsymbol{x}_{k\neg i})$ where $\boldsymbol{x}_{k\neg i} \stackrel{\text{def}}{=} \{x_j\colon z_j = k, j \neq i\}$.
- $p(x_i|\boldsymbol{x}_{k\neg i}) = \iint p\left(x_i \middle| \mu_k, \sigma_k^2\right) p\left(\mu_k, \sigma_k^2 \middle| \boldsymbol{x}_{k\neg i}\right) d\mu_k \, d\sigma_k^2$

1. $p\left(\mu_k, \sigma_k^2 \middle| \boldsymbol{x}_{k\neg i}\right) \propto \mathrm{p}(\boldsymbol{x}_{k\neg i}|\mu_k, \sigma_k^2) p\left(\mu_k, \sigma_k^2\right)$ where $\mathrm{p}(\boldsymbol{x}_{k\neg i}|\mu_k, \sigma_k^2) = \prod_{j=1}^{n_{k\neg i}} \mathcal{N}\left(x_{k\neg i,j} \middle| \mu_{\mathrm{k}}, \sigma_k^2\right)$

- If we want to create new clusters ($n_k = 0$), we must change the prior on μ_k and σ_k^2.
- $p\left(\mu_k, \sigma_k^2\right) = p(\mu_k|\sigma_k) p(\sigma_k) = \mathcal{N}\left(\mu_k \middle| \mu_0, \sigma_k^2/\kappa_0\right) \cdot \mathrm{Inv{-}Gamma}\left(\sigma_k^2 \middle| \alpha_0, \beta_0\right) \stackrel{\text{def}}{=} \mathrm{NIG}\left(\mu_k, \sigma_k^2 \middle| \mu_0, \kappa_0, \alpha_0, \beta_0\right)$
  - $\mu_0 = \left(max(\boldsymbol{x}) + min(\boldsymbol{x})\right)/2 \rightarrow$ middle range.
  - $\kappa_0 = 0.01 \rightarrow$ small value to increase the prior variance (uninformative prior).
  - $\alpha_0 = 2,\ \beta_0 = 4 \rightarrow$ to have good probability in $\sigma_k^2 \in [1,4]$.



MARGINALIZE OVER μ_k AND σ_k^2 (2/3)

- $\text{NIG}\left(\mu_k, \sigma_k^2 \middle| \mu_0, \kappa_0, \alpha_0, \beta_0\right)$ is a **conjugate prior** of $\text{p}(x_{k\neg i}|\mu_k, \sigma_k^2)$.
- $\Rightarrow p\left(\mu_k, \sigma_k^2 \middle| \boldsymbol{x}_{k\neg i}\right) \propto \left(\prod_{j=1}^{n_{k\neg i}} \mathcal{N}\left(x_{k\neg i,j} \middle| \mu_\text{k}, \sigma_k^2\right)\right) \cdot \text{NIG}\left(\mu_k, \sigma_k^2 \middle| \mu_0, \kappa_0, \alpha_0, \beta_0\right) = \text{NIG}\left(\mu_k, \sigma_k^2 \middle| \mu_n, \kappa_n, \alpha_n, \beta_n\right)$
  - $\begin{cases} \alpha_n = \alpha_0 + \frac{n_{k\neg i}}{2}, & \beta_n = \beta_0 + \frac{S_{k\neg i}}{2} + \frac{\kappa_0 n_{k\neg i}}{2\kappa_n}(\overline{x}_{k\neg \text{i}} - \mu_0)^2 \\ \kappa_n = \kappa_0 + n_{k\neg i}, & \mu_n = \frac{\kappa_0 \mu_0 + n_{k\neg i}\overline{x}_{k\neg \text{i}}}{\kappa_0 + n_{k\neg i}} \end{cases}$

2. $p\left(x_i \middle| \mu_k, \sigma_k^2\right) = \mathcal{N}\left(x_i \middle| \mu_\text{k}, \sigma_k^2\right), \qquad p\left(\mu_k, \sigma_k^2 \middle| \boldsymbol{x}_{k\neg i}\right) = \mathcal{N}\left(\mu_k \middle| \mu_n, \sigma_k^2/\kappa_n\right) \cdot \text{Inv−Gamma}\left(\sigma_k^2 \middle| \alpha_n, \beta_\text{n}\right)$

- Let's put the pieces together, isolate the components dependent on μ_k, and apply the convolution of two Gaussians (solution problem 1b: sum of variances).
- $$\begin{aligned} \Rightarrow p(x_i|\boldsymbol{x}_{k\neg i}) &= \int \left(\int \mathcal{N}\left(x_i \middle| \mu_\text{k}, \sigma_k^2\right) \cdot \mathcal{N}\left(\mu_k \middle| \mu_n, \sigma_k^2/\kappa_n\right) d\mu_k\right) \text{Inv−Gamma}\left(\sigma_k^2 \middle| \alpha_n, \beta_\text{n}\right) d\sigma_k^2 = \\ &= \int \mathcal{N}\left(x_i \middle| \mu_n, \sigma_k^2(1 + 1/\kappa_n)\right) \cdot \text{Inv−Gamma}\left(\sigma_k^2 \middle| \alpha_n, \beta_\text{n}\right) d\sigma_k^2 \end{aligned}$$

MARGINALIZE OVER μ_k AND σ_k^2 (3/3)

3. The Inverse-Gamma normalizing constant is: $\int x^{-(\alpha+1)}\, e^{-\beta/x} dx = \Gamma(\alpha)\beta^{-\alpha}$.

- Let's write explicitly the distrubition inside the integral and take out everything that does not depend on σ_k^2.

- $$\begin{aligned} p(x_i|\boldsymbol{x}_{k\neg i}) &= \frac{\beta_n^{\alpha_n}}{\Gamma(\alpha_n)} \frac{1}{\sqrt{2\pi(1+1/\kappa_n)}} \int \left(\sigma_k^2\right)^{-(\alpha_n+1/2+1)} exp\left(-\frac{1}{\sigma_k^2}\left(\beta_n + \frac{(x_i-\mu_n)^2}{2(1+1/\kappa_n)}\right)\right) d\sigma_k^2 = \\ &= \frac{\beta_n^{\alpha_n}}{\Gamma(\alpha_n)} \frac{1}{\sqrt{2\pi(1+1/\kappa_n)}} \Gamma(\alpha_n + 1/2) \left(\beta_n + \frac{(x_i-\mu_n)^2}{2(1+1/\kappa_n)}\right)^{-(\alpha_n+1/2)} \end{aligned}$$

- Let's define $\nu \overset{\text{def}}{=} 2\alpha_n$ and $s^2 \overset{\text{def}}{=} \frac{\beta_n}{\alpha_n}\left(1 + \frac{1}{\kappa_n}\right)$:

$$\Rightarrow p(x_i|\boldsymbol{x}_{k\neg i}) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi\nu s^2}} \left(1 + \frac{(x_i-\mu_n)^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}} \overset{\text{def}}{=} t_\nu(x_i|\mu_n, s)$$

t-Student distribution

COLLAPSED GIBBS SAMPLING ALGORITHM

1. Initialize $\boldsymbol{\theta}^{(0)}$, $\boldsymbol{w}^{(0)}$ and $\boldsymbol{z}^{(0)}$, and choose the hyperparameters $\alpha, \mu_0, \kappa_0, \alpha_0$ and β_0.
2. For iterations $\mathrm{t} = 1,2, \dots, \mathrm{M}$:
   - For each i:
     - Remove point i from the cluster it occupied in the previous iteration $t-1$.
     - Draw $z_i^{(t)} \sim p(z_i = k | z_{\neg i}, \boldsymbol{x}, \alpha) \propto (n_{k \neg i} + \alpha / K) \cdot t_\nu(x_i | \mu_n, s)$.

NON-PARAMETRIC COLLAPSED GIBBS SAMPLING

- Let us assume the number of clusters K is unknown, so we adopt a nonparametric model that permits an **unbounded number of clusters** ($K \to \infty$).
- The only part that depends on K is $p(z_i = k|z_{\neg i}, \alpha)$.
- $p(z_i = k|z_{\neg i}, \alpha) = \lim_{K \to \infty} \frac{n_{k\neg i}+\alpha/K}{N-1+\alpha}$
  - $n_{k\neg i} > 0 \quad \Rightarrow \quad p(z_i = k|z_{\neg i}, \alpha) = \frac{n_{k\neg i}}{N-1+\alpha}$
  - $n_{k\neg i} = 0 \quad \Rightarrow \quad p(z_i = new|z_{\neg i}, \alpha) = 1 - p(z_i = old|z_{\neg i}, \alpha) = 1 - \sum_{k=1}^{K^*} p(z_i = k|z_{\neg i}, \alpha) = 1 - \sum_{k=1}^{K^*} \frac{n_{k\neg i}}{N-1+\alpha} = 1 - \frac{N-1}{N-1+\alpha} = \frac{\alpha}{N-1+\alpha}$
    - K^* denotes the number of occupied clusters, each containing at least one observation.
- $p(z_i = k|z_{\neg i}, \alpha) = \frac{1}{N-1+\alpha} \cdot \begin{cases} n_{k\neg i} & if\ n_{k\neg i} > 0 \\ \alpha & if\ n_{k\neg i} = 0 \end{cases} \Longrightarrow$ **Chinese restaurant process**

NONPARAMETRIC ALGORITHM

1. Initialize $\boldsymbol{\theta}^{(0)}$, $\boldsymbol{w}^{(0)}$ and $\boldsymbol{z}^{(0)}$ and choose the hyperparameters: $\alpha, \mu_0, \kappa_0, \alpha_0$ and β_0.
2. For iterations $\mathrm{t} = 1,2, \dots, \mathrm{M}$:
   - For each i:
     - Remove point i from the cluster it occupied in the previous iteration $t-1$.
     - If the cluster becomes empty, remove it.
     - For each $k = 1, \dots K^*$:compute $p(z_i = k | z_{\neg i}, \boldsymbol{x}, \alpha) \propto n_{k \neg i} \cdot t_\nu(x_i | \mu_n, s) \stackrel{\text{def}}{=} q_k$.
     - Compute $p(z_i = new | z_{\neg i}, \boldsymbol{x}, \alpha) \propto \alpha \cdot t_\nu(x_i | \mu_0, s) \stackrel{\text{def}}{=} q_{\text{new}}$.
     - Normalize: $p_k = q_k / \left(q_{new} + \sum_{k=1}^{K^*} q_k\right)$ and $p_{\text{new}} = q_{\text{new}} / \left(q_{new} + \sum_{k=1}^{K^*} q_k\right)$
     - Draw $z_i^{(t)}$ with `numpy.random.choice` using probabilities $p = (p_1, \dots, p_{K^*}, p_{new})$
     - If $z_i^{(t)} = new$, create a new cluster and assign i to it;
       otherwise assign i to the selected existing cluster.

MULTIDIMENSIONAL GENERALIZATION (1/3)

- The data-dependent term is $p(x_i|\boldsymbol{x_{k\neg i}})$, where $x_i = (x_{i_1}, \dots, x_{i_d})$, $x_i \in \mathbb{R}^d$.
- $p(x_i|\boldsymbol{x}_{k\neg i}) = \iint p(x_i|\mu_k, \Sigma_k) p(\mu_k, \Sigma_k|\boldsymbol{x}_{k\neg i}) d\mu_k \, d\Sigma_k$
  - Here $\mu_k = (\mathrm{\mu_{k_1}}, \dots, \mathrm{\mu_{k_d}})$ and Σ_k is the $d \times d$ covariance matrix.

1. $p(\mu_k, \Sigma_k|\boldsymbol{x}_{k\neg i}) \propto p(\boldsymbol{x}_{k\neg i}|\mu_k, \Sigma_k) p(\mu_k, \Sigma_k)$

  - $\mathrm{p}(\boldsymbol{x}_{k\neg i}|\mu_k, \Sigma_k) = \prod_{j=1}^{n_{k\neg i}} \mathcal{N}\left(x_{k\neg i,j} \middle| \mu_\mathrm{k}, \Sigma_k\right)$, the likelihood is a products of **multivariate Gaussian**.

- $p(\mu_k, \Sigma_k) = p(\mu_k|\Sigma_k) p(\Sigma_k) = \mathcal{N}(\mu_k|\mu_0, \Sigma_k/\kappa_0) \cdot \text{Inv}-\text{Wishart}(\Sigma_k|\nu_0, \Lambda_0) \stackrel{\text{def}}{=} \text{NIW}(\mu_k, \Sigma_k|\mu_0, \kappa_0, \nu_0, \Lambda_0)$
  - $\textbf{Inv}-\textbf{Wishart}(\boldsymbol{\Sigma_k}|\boldsymbol{\nu_0}, \boldsymbol{\Lambda_0})$ is the multivariate analogue of the Inv-Gamma $\left(\sigma_k^2 \middle| \alpha_0, \beta_0\right)$.
  - If $d = 1 \Rightarrow \nu_0 = 2\alpha_0$, and $\Lambda_0 = 2\beta_0$. For consistency, we choose $\nu_0 = 4$, and $\Lambda_0 = 8 \cdot I_{d\times d}$.
  - $\kappa_0 = 0.01 \rightarrow$ small value to increase the prior covariance (uninformative prior).
  - $\mu_0 = \left\{\left(max\left(x_{i_h}\right) + min\left(x_{i_h}\right)\right)/2\right\}_{h=1}^{d} \rightarrow$ middle range.

MULTIDIMENSIONAL GENERALIZATION (2/3)

- $\mathrm{NIW}(\mu_k, \Sigma_k | \mu_0, \kappa_0, \nu_0, \Lambda_0)$ is a **conjugate prior** of $\mathrm{p}(x_{k\neg i} | \mu_k, \Sigma_k)$.
- $\Rightarrow p(\mu_k, \Sigma_k | \boldsymbol{x}_{k\neg i}) \propto \left(\prod_{j=1}^{n_{k\neg i}} \mathcal{N}\left(x_{k\neg i,j} \middle| \mu_\mathrm{k}, \Sigma_k\right)\right) \cdot \mathrm{NIG}(\mu_k, \Sigma_k | \mu_0, \kappa_0, \nu_0, \Lambda_0) = \mathrm{NIG}(\mu_k, \Sigma_k | \mu_n, \kappa_n, \nu_n, \Lambda_n)$
  - $\begin{cases} \nu_n = \nu_0 + n_{k\neg i}, \qquad \Lambda_n = \Lambda_0 + S_{k\neg i} + \frac{\kappa_0 n_{k\neg i}}{\kappa_n}(\overline{x}_{k\neg \mathrm{i}} - \mu_0)(\overline{x}_{k\neg \mathrm{i}} - \mu_0)^\intercal \\ \kappa_n = \kappa_0 + n_{k\neg i}, \qquad \mu_n = \frac{\kappa_0 \mu_0 + n_{k\neg i}\overline{x}_{k\neg \mathrm{i}}}{\kappa_0 + n_{k\neg i}} \end{cases}$

2. $p(x_i | \mu_k, \Sigma_k) = \mathcal{N}(x_i | \mu_\mathrm{k}, \Sigma_k), \qquad p(\mu_k, \Sigma_k | \boldsymbol{x}_{k\neg i}) = \mathcal{N}(\mu_k | \mu_n, \Sigma_k / \kappa_n) \cdot \mathrm{Inv{-}Wishart}(\Sigma_k | \nu_n, \Lambda_n)$

- Let's put the pieces together, isolate the components dependent on μ_k, and apply the convolution of two Gaussians.
- $$\begin{aligned} \Rightarrow p(x_i | \boldsymbol{x}_{k\neg i}) &= \int \left(\int \mathcal{N}(x_i | \mu_\mathrm{k}, \Sigma_k) \cdot \mathcal{N}(\mu_k | \mu_n, \Sigma_k / \kappa_n) d\mu_k\right) \mathrm{Inv{-}Wishart}(\Sigma_k | \nu_n, \Lambda_n) d\Sigma_k = \\ &= \int \mathcal{N}(x_i | \mu_n, \Sigma_k(1 + 1/\kappa_n)) \cdot \mathrm{Inv{-}Wishart}(\Sigma_k | \nu_n, \Lambda_n) d\Sigma_k \end{aligned}$$

MULTIDIMENSIONAL GENERALIZATION (3/3)

3. The Inverse-Wishart normalizing constant is: $\int |\Sigma|^{-(\nu+d+1)/2}\, e^{-tr(\Lambda\Sigma^{-1})/2} d\Sigma = \left(2^{\nu d/2}\Gamma_d(\nu/2)\right)/|\Lambda|^{\nu/2}$.

- Let's write explicitly the distrubition inside and define $c \overset{\text{def}}{=} \kappa_n/(\kappa_n+1)$ and $u \overset{\text{def}}{=} x_i - \mu_n$.

$$
\begin{aligned}
p(x_i|\boldsymbol{x}_{k\neg i}) &= (2\pi)^{-\frac{d}{2}} c^{\frac{d}{2}} \frac{|\Lambda_n|^{\nu_n/2}}{2^{\nu_n d/2}\Gamma_d(\nu_n/2)} \int |\Sigma_k|^{-\frac{1}{2}} |\Sigma_k|^{-\frac{\nu_n+d+1}{2}} e^{-\frac{1}{2}u^\top \Sigma_k^{-1} u} e^{-\frac{1}{2}tr(\Lambda_n \Sigma_k^{-1})} d\Sigma_k = \\
&= (2\pi)^{-\frac{d}{2}} c^{\frac{d}{2}} \frac{|\Lambda_n|^{\nu_n/2}}{2^{\nu_n d/2}\Gamma_d(\nu_n/2)} \int |\Sigma_k|^{-\frac{(\nu_n+1)+d+1}{2}} e^{-\frac{1}{2}tr\left((cuu^\top+\Lambda_n)\Sigma_k^{-1}\right)} d\Sigma_k \\
&= \pi^{-\frac{d}{2}} c^{\frac{d}{2}} \frac{\Gamma_d\left((\nu_n+1)/2\right)}{\Gamma_d(\nu_n/2)} |\Lambda_n|^{-1/2} (1 + cu^\top \Lambda_n^{-1} u)^{-(\nu_n+d)/2}
\end{aligned}
$$

- Let's define $\nu_t \overset{\text{def}}{=} \nu_n - d + 1$ and $\Sigma_t \overset{\text{def}}{=} \frac{\kappa_n+1}{\kappa_n \nu_t}\Lambda_n$:

$$
\Rightarrow p(x_i|\boldsymbol{x}_{k\neg i}) = \frac{\Gamma\left(\frac{\nu_t+1}{2}\right)}{\Gamma\left(\frac{\nu_t}{2}\right)(\pi\nu_t)^{d/2}|\Sigma_t|^{1/2}} \left(1 + \frac{1}{\nu_t}(x_i-\mu_n)^\top \Sigma_t^{-1}(x_i-\mu_n)\right)^{-\frac{\nu_t+d}{2}} \overset{\text{def}}{=} t_{\nu_t}(x_i|\mu_n, \Sigma_t)
$$

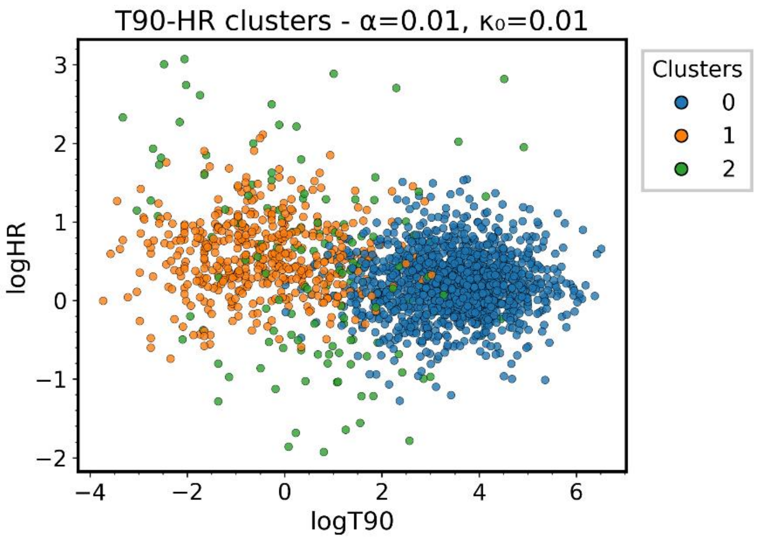
Multivariate t-distribution

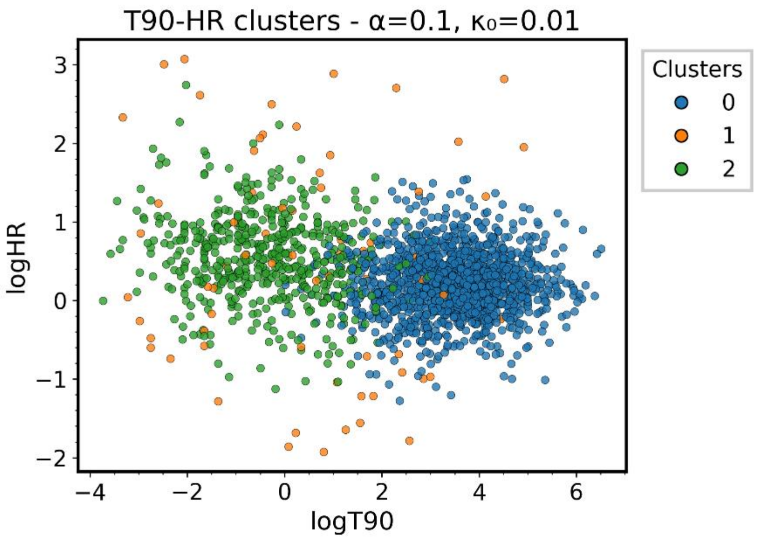
MULTIDIMENSIONAL ALGORITHM

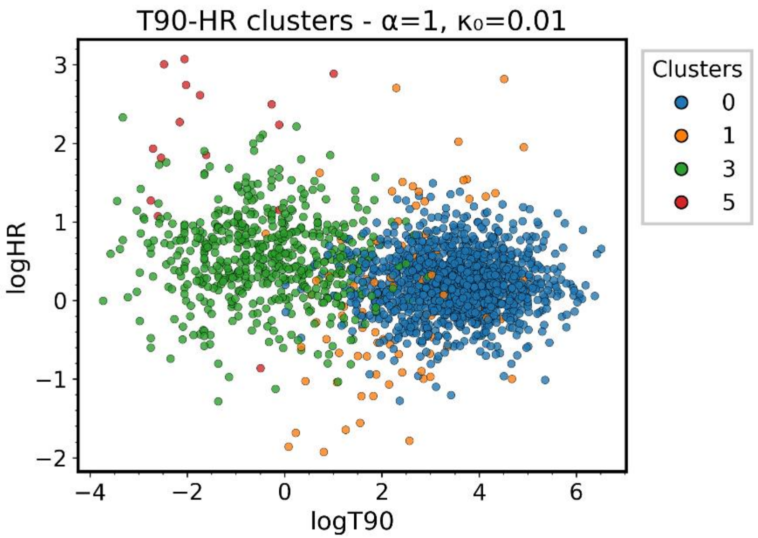
1. Initialize $\boldsymbol{\theta}^{(0)}$, $\boldsymbol{w}^{(0)}$ and $\boldsymbol{z}^{(0)}$ and choose the hyperparameters: $\alpha, \mu_0, \kappa_0, \nu_0$ and Σ_0.
2. For iterations $\mathrm{t} = 1,2, \dots, \mathrm{M}$:
   - For each i:
     - Remove point i from the cluster it occupied in the previous iteration $t-1$.
     - If the cluster becomes empty, remove it.
     - For each $k = 1, \dots K^*$: compute $p(z_i = k | z_{\neg i}, \boldsymbol{x}, \alpha) \propto n_{k \neg i} \cdot t_{\nu_t}(x_i | \mu_n, \Sigma_t) \stackrel{\text{def}}{=} q_k$.
     - Compute $p(z_i = new | z_{\neg i}, \boldsymbol{x}, \alpha) \propto \alpha \cdot t_{\nu_t}(x_i | \mu_0, \Sigma_t) \stackrel{\text{def}}{=} q_{\text{new}}$.
     - Normalize: $p_k = q_k / \left(q_{new} + \sum_{k=1}^{K^*} q_k\right)$ and $p_{\text{new}} = q_{\text{new}} / \left(q_{new} + \sum_{k=1}^{K^*} q_k\right)$
     - Draw $z_i^{(t)}$ with `numpy.random.choice` using probabilities $p = (p_1, \dots, p_{K^*}, p_{new})$
     - If $z_i^{(t)} = new$, create a new cluster and assign i to it;
       otherwise assign i to the selected existing cluster.

VARIABLE α RESULTS.

#Iterations = 1000, #Initial clusters=2







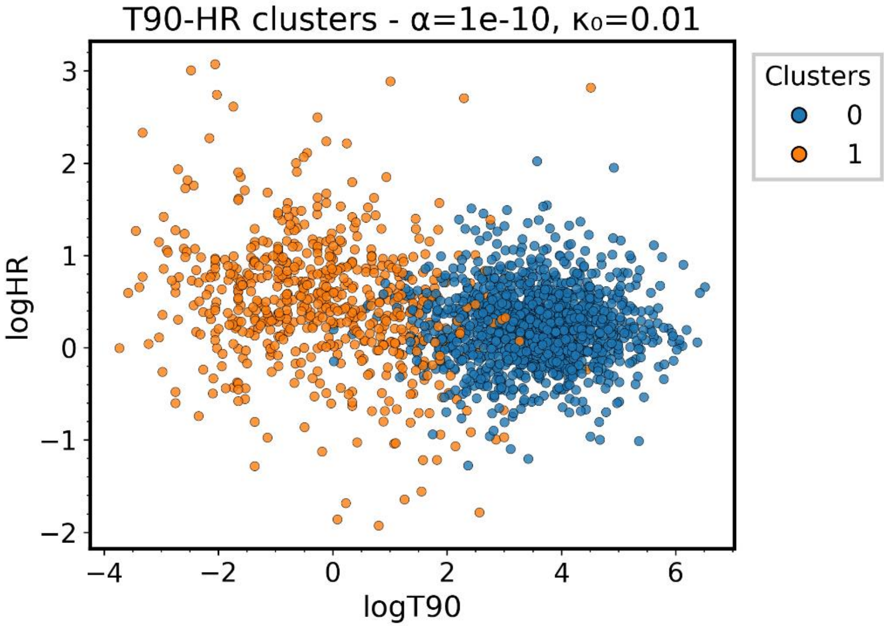
- $\mu_0 = [3.51, 0.212]$, $\sigma_0 = [1.02, 0.395]$
- $\mu_1 = [-0.467, 0.566]$, $\sigma_1 = [1.27, 0.482]$
- $\mu_2 = [-0.541, 0.465]$, $\sigma_2 = [1.68, 1.16]$

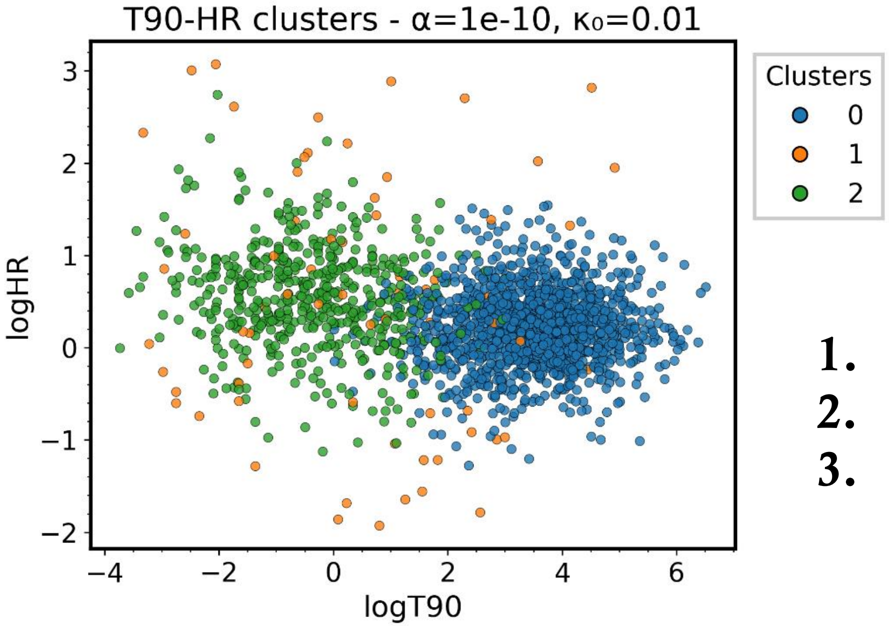
- $\mu_0 = [3.50, 0.216]$, $\sigma_0 = [1.02, 0.399]$
- $\mu_1 = [0.458, 0.480]$, $\sigma_1 = [2.06, 1.34]$
- $\mu_2 = [-0.373, 0.547]$, $\sigma_2 = [1.27, 0.554]$

- $\mu_0 = [3.53, 0.216]$, $\sigma_0 = [1.01, 0.38]$
- $\mu_1 = [2.33, 0.075]$, $\sigma_1 = [1.19, 1.00]$
- $\mu_3 = [-0.395, 0.544]$, $\sigma_3 = [1.28, 0.551]$
- $\mu_5 = [-1.51, 1.20]$, $\sigma_5 = [1.16, 0.986]$

FIXED COMPONENTS.

#Iterations = 1000





1. **Short GRB → Hard GRB**
2. **Long GRB → Soft GRB**
3. **Outliers**

- #Initial clusters = 2
- $\mu_0 = [3.55, 0.228]$, $\sigma_0 = [0.987, 0.402]$
- $\mu_1 = [-0.174, 0.495]$, $\sigma_1 = [1.43, 0.692]$

- #Initial clusters = 3
- $\mu_0 = [3.50, 0.216]$, $\sigma_0 = [1.02, 0.399]$
- $\mu_1 = [0.458, 0.480]$, $\sigma_1 = [2.06, 1.34]$
- $\mu_2 = [-0.373, 0.547]$, $\sigma_2 = [1.27, 0.554]$

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